HIERARCHICAL HYPERBOLICITY OF COXETER GROUPS

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Abstract. We show that for every finite rank Coxeter system, the Coxeter group is hierarchically hyperbolic relative to its nonhyperoctahedral irreducible higher rank affine parabolic subgroups. This is sharp—the Coxeter group admits a hierarchically hyperbolic group structure if and only if it has no such parabolic subgroups, and we give a complete diagrammatic characterization of when this is true. Moreover, this condition is equivalent to the existence of a coarsely equivariant coarse median on the Coxeter group.

We also show that the Coxeter group is virtually a colorable hierarchically hyperbolic group. This is achieved by passing to a congruence subgroup of the Tits representation.

As consequences, the Davis complex is a hierarchically hyperbolic space and the group admits a coarse median of rank equal to its algebraic rank. Furthermore, the Coxeter group is quasimedian quasiisometric to a finite dimensional CAT(0) cube complex and is virtually coarsely injective.

Two appendices prove results of independent interest that are not Coxeter-specific. The first promotes a relative HHS structure with HHS peripheral coordinate spaces to an HHS structure. The second shows that a virtually Abelian group admits a coarsely equivariant coarse median precisely when it is hyperoctahedral.

1. Introduction

Coxeter groups have a rich geometry arising from their wall structure and the structure of their parabolic subgroups. In a previous paper [17] we showed that hyperbolic aspects of this geometry can be understood in close analogy to those of the mapping class group of a hyperbolic surface. By coning off the nonhyperbolic regions in the Davis complex $\Sigma = \Sigma(W, S)$ of a finite rank Coxeter system (W, S), we produced a hyperbolic space $\widehat{\Sigma}$ playing the role of the curve graph of the surface: it recognizes the Morse quasigeodesics and stable subspaces of the Davis complex, and provides the largest acylindrically hyperbolic structure on the Coxeter group. However, coning off the nonhyperbolic regions collapses them to bounded diameter, effectively discarding the geometry of unbounded regions in Σ. The purpose of this paper is to recover that missing geometry, in a way that further strengthens the mapping class group analogy.

For mapping class groups, a vertex in the curve graph corresponds to a curve in the surface. Cutting the surface along that curve gives a subsurface whose components have strictly lower complexity and are equipped with their own mapping class groups and curve graphs. This leads to an inductive, hierarchical description of mapping class group geometry that uses subsurface projection to recover the geometry lost in passing to the curve graph [29, 30]. For a Coxeter group, a cone vertex of $\widehat{\Sigma}$ is associated to a parabolic subcomplex of Σ. This is the Davis complex of a parabolic subgroup of (W, S), whose irreducible factors are Coxeter groups of strictly lower rank, suggesting an analogous hierarchical induction to recover the geometry lost in passing to $\widehat{\Sigma}$. The notion of a hierarchically hyperbolic space formalizes these hierarchical constructions.

A *hierarchically hyperbolic space* (HHS) is a quasigeodesic metric space described by a system of hyperbolic coordinate spaces that are indexed by hierarchically related "domains". A set of axioms governs the interactions between domains, their coordinate spaces, and projections to the coordinate spaces, in such a way that consistent coordinate data can be coarsely realized by a point in the original space. Moreover, a distance formula recovers the original metric, up to quasiisometry, from the sum of sufficiently large coordinate distances. A *hierarchically hyperbolic group* (HHG) is a group with an HHS structure on its Cayley graph that has additional equivariance properties. HHSs were introduced to provide a common framework for studying the geometry of mapping class groups and cocompactly cubulated groups [5], thus providing a bridge to transfer results from one setting to the other.

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A *relative HHS/HHG* structure is a relaxation of the requirements that allows some minimal domains to be designated "peripheral". The coordinate space of a peripheral domain is not required to be hyperbolic.

Not all Coxeter groups admit an HHG structure [38, 4]. The attempted analogy with mapping class groups breaks down when we encounter irreducible higher rank affine parabolic subgroups, which are not hyperbolic, but also do not decompose further by induction on parabolic subgroups. We build a relative HHG structure by stopping when we reach such parabolic subgroups:

Theorem 1.1. *Let (W, S) be a finite rank Coxeter system. Then W is hierarchically hyperbolic relative to the irreducible higher rank affine parabolic subgroups of (W, S).*

The relative HHG structure we define in Section 4 is naturally related to the Coxeter structure: the domains are certain of the parabolic subgroups, nesting of domains is subgroup containment, orthogonality of domains is commutation of subgroups, and coordinate projections are induced by gate projections in Σ. This parallels the mapping class group structure, in which domains are subsurfaces, nesting is containment of subsurfaces, and orthogonality is disjointness of subsurfaces.

The natural Coxeter/parabolic induction stops at irreducible higher rank affine parabolics, so to proceed further we equip these parabolic subgroups with HHS structures from their crystallographic geometries. There is a general scheme, see Appendix A, for promoting a relative HHS structure to an absolute HHS structure by replacing each peripheral domain with an HHS structure on its coordinate space. Making the result an HHG structure requires compatibility with the group action. In Section 5 we consider two ways to arrange such compatibility: restrict the types of affine parabolic subgroups that occur, or pass to a finite index subgroup.

A crystallographic group acts on a Euclidean space with a finite-index translation subgroup and a finite linear part, its "point group". The desired group compatibility is possible precisely when the parabolic subgroup is *hyperoctahedral*: there exists a choice of basis that is permuted, up to sign, by the point group action. We resolve the hyperoctahedral irreducible higher rank affine parabolic subgroups to get a refined relative HHG structure on all of W:

Theorem 1.2. *Let (W, S) be a finite rank Coxeter system. Then W is an HHG relative to the nonhyperoctahedral irreducible higher rank affine parabolic subgroups of (W, S). In particular, W is an HHG if it has no nonhyperoctahedral irreducible higher rank affine parabolic subgroups.*

In fact, Theorem 1.2 is sharp, and we give a complete, diagrammatic characterization of being an HHG for Coxeter groups. The argument hinges on a Coxeter nonhyperoctahedral escalation lemma, Lemma 5.4, and Theorem B.5, which characterizes the existence of coarsely equivariant coarse medians on virtually Abelian groups: one exists if and only if the group is cocompactly cubulable, which is equivalent to hyperoctahedrality. We use *coarse median* in the sense of Bowditch [11], see Definition B.1 and Definition B.2.

Theorem 1.3. *Let (W, S) be a finite rank Coxeter system. The following are equivalent:*

1. *W admits an HHG structure.*
2. *W admits a coarsely equivariant coarse median.*
3. *(W, S) has no nonhyperoctahedral irreducible higher rank affine parabolic subgroup.*
4. *The Coxeter diagram of (W, S) contains no induced subdiagram of irreducible affine type $\widetilde{A}_2$, $\widetilde{A}_n$ for $n \geq 4$, $\widetilde{E}_6$, $\widetilde{E}_7$, $\widetilde{E}_8$, $\widetilde{F}_4$, or $\widetilde{G}_2$.*

A Coxeter system is *affine-free* if (W, S) has no irreducible higher rank affine parabolic subgroups. Niblo and Reeves [33] built a CAT(0) cube complex X_{NR} whose wall structure is W–equivariantly isomorphic to that of the Davis complex of (W, S). The action $W \curvearrowright X_{\text{NR}}$ is cocompact if and only if (W, S) is affine-free [16]. In particular, if (W, S) is affine-free then one can pull back the cubical median from X_{NR} to get a coarsely equivariant coarse median on W. Fioravanti, Levcovitz, and Sageev, [20, Section 1.3], say that when a Coxeter group W is not affine-free it is not known if W admits a coarsely equivariant coarse median. Theorem 1.3 determines precisely when such a median exists.

Azuelos and Hagen conjecture that W is an HHG if and only if it is cocompactly cubulated [4, Conjecture 1.5]; see also [2]. Every cocompactly cubulated group admits a coarsely equivariant coarse median, so Theorem 1.3 establishes one direction of their conjecture. For the converse, if W is an HHG then it has no parabolics of the affine types forbidden by Theorem 1.3, but we do not know if that is a sufficient condition for the existence of a cocompact cubulation. The obstruction to cocompact cubulation of Munro and Petyt [32], using richly branching flats of maximal dimension, cannot apply to Coxeter groups: it obstructs the existence

of any coarse median, not just those that are coarsely equivariant, but we show in Theorem 1.6 that Coxeter groups always admit a coarse median.

The other route to an HHG is by passing to a finite index subgroup of W. We show that for any odd prime p, the p–principal congruence subgroup of the Tits representation of W is a finite index subgroup $K_{(p)} \leq W$ with the property that the action of the $K_{(p)}$–stabilizer of every irreducible higher rank affine parabolic subgroup factors through its translation subgroup. We then get an HHG structure on $K_{(p)}$ (in fact, a colorable HHG structure, see Definition 5.1):

Theorem 1.4. *Let (W, S) be a finite rank Coxeter system. Then W is virtually a colorable hierarchically hyperbolic group; that is, W has a finite index subgroup that is a colorable HHG.*

Since being an HHS is preserved by quasiisometry, it follows that W and Σ have HHS structures:

Corollary 1.5. *The Davis complex of a finite rank Coxeter system is an HHS.*

Theorem 1.4 and Corollary 1.5 answer questions from the hierarchical hyperbolicity problem list [2] on whether Coxeter groups are always virtual HHGs and whether Davis complexes are HHSs.

New Coxeter group results emerge from the abstract HHG framework:

Theorem 1.6. *Every finitely generated Coxeter group has a coarse median of rank equal to its algebraic rank.*

The *algebraic rank* is the maximal rank of a free Abelian subgroup. For Coxeter groups, the algebraic rank is easily determined directly from the Coxeter diagram [28, Theorem 6.8.3]. Theorem 1.6 is optimal in terms of rank: since W is a CAT(0) group, Abelian subgroups are undistorted, so by [11, Corollary 2.4], the algebraic rank is a lower bound for the rank of any coarse median on W. For HHGs, the rank of the HHS coarse median is equal to the algebraic rank, [7, Theorem 1.15 and Corollary 2.15] and [4, Theorem 7.14]. Algebraic rank is invariant upon passage to a finite index subgroup, so Theorem 1.6 follows.

Theorem 1.6 says in particular that every finitely generated Coxeter group admits a coarse median. This is a nontrivial result in itself: Haettel [21] shows that many thick affine buildings have no coarse medians at all, so the existence of a coarse median does not follow just from the fact that Σ is a CAT(0) polyhedral complex.

A theorem of Petyt, [37, Theorem B], turns the "colorable" conclusion of Theorem 1.4 into a cubical model:

Theorem 1.7. *Every finitely generated Coxeter group is quasiisometric to a finite dimensional CAT(0) cube complex via a quasiisometry that coarsely matches the cubical median with the coarse median on the group coming from its virtual HHG structure.*

Haglund and Wise [24] proved that there is a finite index subgroup of W whose action on $X_{\mathbb{NR}}$ is "special". Because of the lack of cocompactness in the non-affine-free case, neither Niblo-Reeves nor Haglund-Wise provides a quasiisometry to a CAT(0) cube complex for arbitrary finitely generated Coxeter groups.

A further consequence of the virtual HHG structure concerns actions on injective metric spaces (also known as hyperconvex spaces). A geodesic metric space is *injective* if closed balls have the Helly property: every family of pairwise intersecting closed balls has nonempty intersection. Theorem 1.4 combined with a result of Haettel, Hoda, and Petyt [22, Lemma 3.10] gives:

Theorem 1.8. *Every finitely generated Coxeter group is* virtually coarsely injective*: it has a finite index subgroup that acts properly and coboundedly by isometries on an injective metric space.*

Passing to a finite index subgroup is necessary, in general, because Hoda [25, Theorem 7.1] shows that a crystallographic group is coarsely injective if and only if it is hyperoctahedral, and Theorem 2.11 provides examples of nonhyperoctahedral crystallographic Coxeter groups.

On a finite dimensional CAT(0) cube complex, the path metric induced by taking the ℓ^∞ metric on each cube gives an injective space [12, Lemma 7.5 and Theorem 7.8], so this metric on the cube complex of Theorem 1.7 gives a nonequivariant injective model for a Coxeter group, whereas Theorem 1.8 gives an equivariant injective model for a finite index subgroup.

2. Preliminaries

2.1. **Coxeter groups.** We provide a quick recap. For further background and justifications, see [17].

A *Coxeter system* (W, S) is a Coxeter group W and a fundamental generating set S of involutions such that if m_{st} is the order of the element st, for $s, t \in S$, then W is the group with presentation:

$$W = \langle S \mid (st)^{m_{st}} \text{ for all } s, t \in S \text{ with } m_{st} < \infty \rangle$$

A *special subgroup* of (W, S) is a subgroup W_T generated by a subset $T \subset S$. A *parabolic subgroup* of (W, S) is a conjugate of a special subgroup. If $T \subset S$ then (W_T, T) is a Coxeter system, [18, Theorem 4.1.6(i)]. By [18, Lemma 5.3.6], the intersection of two parabolic subgroups is a parabolic subgroup. Since S is finite, it follows that given a set of elements $A \subset W$, there is a well-defined smallest parabolic subgroup containing A, called its *parabolic closure*, and denoted $\mathrm{Pc}(A)$. Moreover, from the proof of [18, Lemma 5.3.6] and the subsequent remark, it follows that for every parabolic subgroup P of (W, S), the intersection $P \cap W_T$ is a parabolic subgroup of both (W, S) and (W_T, T). In particular, the parabolic subgroups of (W_T, T) are precisely the parabolic subgroups of (W, S) that are contained in W_T.

The Coxeter presentation can also be encoded by a *Coxeter diagram*, which is a graph with vertex set S, and an edge labelled m_{st} between vertices $s, t \in S$ when $m_{st} > 2$. If the Coxeter diagram is connected then the Coxeter system is called *irreducible*. The *canonical product decomposition* of (W, S) is $W = W_{S_1} \times \cdots \times W_{S_n}$, where the S_i are the vertex sets of the connected components of the Coxeter diagram.

For $T \subset S$, let $T^\perp := \{s \in S \smallsetminus T \mid \forall t \in T,\ st = ts\}$.

Coxeter groups are called *spherical* or *affine* if they act as geometric reflection groups on a sphere or Euclidean space, respectively. A Coxeter group is spherical if and only if it is finite. Since parabolic subgroups are themselves Coxeter groups, this terminology is also used to describe parabolic subgroups. By extension, a subset $T \subset S$ is called spherical, affine, irreducible, etc, according to the corresponding property for W_T.

The *rank* of a Coxeter system is $|S|$. An irreducible *higher rank* affine parabolic subgroup is one of rank at least 3, which is equivalent to the corresponding Euclidean space having dimension at least 2.

A Coxeter system is *wide* if its canonical product decomposition into irreducible factors either contains at least two nonspherical factors or contains an irreducible higher rank affine factor. A nonspherical irreducible parabolic subgroup is *pre-wide* if its normalizer is wide.

Lemma 2.1 ([17, Lemma 5.12])**.** *If P is a nonspherical irreducible parabolic subgroup that is contained in a wide parabolic subgroup, then P is pre-wide.*

A finite rank Coxeter system (W, S) admits a geometric action on a piecewise Euclidean, CAT(0) polyhedral cell complex Σ, its *Davis complex*. By fixing a base vertex, the 1–skeleton $\Sigma^{(1)}$ is identified with the Cayley graph of W with respect to S. The path metric on $\Sigma^{(1)}$ is denoted d_1. For $T \subset S$, the subcomplex $w\Sigma_T$ of Σ spanned by the coset wW_T is called a *parabolic subcomplex*. Its stabilizer $\mathrm{Stab}(w\Sigma_T) = wW_Tw^{-1}$ is a parabolic subgroup. When P is a parabolic subgroup we write $\Upsilon \in \mathrm{Stab}^{-1}(P)$ to mean that Υ is a parabolic subcomplex with $\mathrm{Stab}(\Upsilon) = P$, and we say Υ is a *P–fiber*. When $\Xi \subset \Upsilon$ are parabolic subcomplexes, $\mathrm{Stab}(\Xi) \leq \mathrm{Stab}(\Upsilon)$. Conversely if $P \subset Q$ are parabolic subgroups, then for every $\Xi \in \mathrm{Stab}^{-1}(Q)$ there exists $\Upsilon \in \mathrm{Stab}^{-1}(P)$ with $\Upsilon \subset \Xi$.

If Υ is a parabolic subcomplex of Σ then $\Upsilon^{(1)}$ is a convex subgraph of $\Sigma^{(1)}$ and closest point projection $\mathfrak{p}_\Upsilon \colon \Sigma^{(0)} \to \Upsilon^{(0)}$ is a *gate map*, which means that for all $x \in \Sigma^{(0)}$ and $y \in \Upsilon^{(0)}$ we have:

$$d_1(x, y) = d_1(x, \mathfrak{p}_\Upsilon(x)) + d_1(\mathfrak{p}_\Upsilon(x), y)$$

It follows easily that $\mathfrak{p}_\Upsilon$ is 1–Lipschitz on $\Sigma^{(0)}$ and restricts to the identity on $\Upsilon^{(0)}$. For $X \subset \Sigma$, the notation $\mathfrak{p}_\Upsilon(X)$ means $\bigcup_{x \in X \cap \Sigma^{(0)}} \mathfrak{p}_\Upsilon(x)$. We recall some further properties of gate maps to parabolic subcomplexes:

Lemma 2.2 ([17, Proclamations 2.42–2.47])**.** *For all $w \in W$, $x \in \Sigma^{(0)}$, and parabolic subcomplexes Υ and Ξ:*

(1) $\Upsilon \subset \Xi \implies \mathfrak{p}_\Upsilon = \mathfrak{p}_\Upsilon \circ \mathfrak{p}_\Xi$
(2) A wall separates x from $\mathfrak{p}_\Upsilon(x)$ if and only if it separates x from Υ.
(3) $\mathfrak{p}_{w\Upsilon}(x) = w\mathfrak{p}_\Upsilon(w^{-1}x)$
(4) If $T \subset S$ and $z \in W_{T^\perp}$ then $\mathfrak{p}_{w\Sigma_T}(wzw^{-1}x) = \mathfrak{p}_{w\Sigma_T}(x)$.
(5) $\mathfrak{p}_\Upsilon(\Xi)$ *spans a parabolic subcomplex of* Υ.
(6) $\mathrm{Stab}(\mathfrak{p}_\Xi(\Upsilon)) = \mathrm{Stab}(\mathfrak{p}_\Upsilon(\Xi)) = \mathrm{Stab}(\Upsilon) \cap \mathrm{Stab}(\Xi)$

A conjugate of a generator is called a *reflection*. It acts on Σ as a reflection, a CAT(0)–isometric involution fixing a *wall* pointwise and exchanging the two complementary components of the wall, which are called *open halfspaces*. Distinct walls of Σ are either disjoint or intersect transversely, and their corresponding

reflections generate an infinite or finite dihedral group, respectively. For $x, y \in \Sigma^{(0)}$, let $\mathcal{M}(x,y)$ be the set of walls separating x from y; that is, the walls such that x is in one open halfspace and y is in the other. The combinatorial distance $d_1(x,y)$ from x to y in $\Sigma^{(1)}$ is exactly $|\mathcal{M}(x,y)|$, and for every combinatorial geodesic γ from x to y, the edges of γ are in bijection with walls of $\mathcal{M}(x,y)$ by matching edge e with the unique wall M separating the vertices of e. We say M *cuts* e. Similarly, we say M *cuts* a parabolic subcomplex Υ if it cuts some edge of Υ. A wall M cuts Υ if and only if the reflection in M belongs to the stabilizer of Υ.

A *chain* is an ordered collection of distinct walls $\mathcal{K} = \{K_i\}_{i\in I}$ for $I \subset \mathbb{Z}$ such that there is a choice of open halfspaces K_i^- and K_i^+ of wall K_i such that $i < j$ in I implies $K_i^- \subset K_j^-$. Its length is the number of walls it contains. All the chains in this paper have finite length. In [17] we defined a *Caprace constant*, ℓ, depending on the Coxeter system. The *interior* $\mathring{\mathcal{K}}$ of a chain is the subchain obtained from $\mathcal{K}$ by discarding the $(\ell-1)$–outermost walls on each end, so if $\mathcal{K} = \{K_i\}_{1\le i\le n}$, then $\mathring{\mathcal{K}} = \{K_i\}_{\ell\le i\le n+1-\ell}$. A *grid* is a pair $(\mathcal{V},\mathcal{H})$ such that $\mathcal{V}$ and $\mathcal{H}$ are chains containing at least two walls each, such that every wall of $\mathcal{V}$ is transverse to every wall of $\mathcal{H}$. When $\mathcal{M}$ is a set of walls, let $\mathrm{Pc}(\mathcal{M})$ be the parabolic closure of the corresponding set of reflections. When $\mathcal{V}$ is a chain of length at least 2, $\mathrm{Pc}(\mathcal{V})$ is a nonspherical irreducible parabolic subgroup.

A *reflection subgroup* is a subgroup generated by reflections. The following is well-known:

Lemma 2.3. *A reflection subgroup of a direct product of special subgroups splits as a direct product of reflection subgroups of the factors. Consequently, if P is an irreducible parabolic subgroup and $Q = Q_1 \times Q_2$ is parabolic subgroup that splits as a product of parabolic subgroups, then:*

$$P \le Q \iff (P \le Q_1) \vee (P \le Q_2)$$

2.2. **Essential parabolic subgroups.** In [17] we used known results about parabolic centralizers and normalizers to say that a nonspherical irreducible parabolic subgroup P has an "orthogonal complement" $P^\perp$ that is parabolic, equal to the centralizer $Z_W(P)$ of P, and such that $P \times P^\perp$ is parabolic and equal to the normalizer $N_W(P)$ of P. Then we used these facts to understand the structure of parabolic subcomplexes stabilized by P. Now, following Caprace and Marquis [14] and Caprace, Marquis, and Reid [15], we extend these results to a wider class of parabolic subgroups.

Definition 2.4. A parabolic subgroup P of (W,S) has a canonical decomposition as a product of irreducible factors, $P = Q_1 \times \cdots \times Q_m \times F_1 \times \cdots \times F_n$, where the Q_i are nonspherical and the F_j are spherical. Call $P^{ess} := Q_1 \times \cdots \times Q_m$ the *essential part* of P. Call P *essential* if $P = P^{ess}$.

Lemma 2.5. *If $P \le Q$ are parabolic subgroups of (W,S) then $P^{ess} \le Q^{ess}$.*

Proof. Apply Lemma 2.3 to the canonical product decompositions of P and Q. □

Lemma 2.6 ([14, Lemma 2.1], expanded proof as [15, Lemma 3.3])**.** *If P is an essential parabolic subgroup of (W,S) then there exist $w \in W$ and a unique $T \subset S$ such that $P = wW_Tw^{-1}$, $Z_W(P) = wW_{T^\perp}w^{-1}$, and $N_W(P) = wW_{T\sqcup T^\perp}w^{-1}$, so $P \times Z_W(P) = N_W(P)$ and both $Z_W(P)$ and $N_W(P)$ are parabolic subgroups.*

Definition 2.7. If P is an essential parabolic subgroup of (W,S), define $P^\perp := Z_W(P)$.

This extends the $P^\perp$ notation for P nonspherical and irreducible from [17]. In addition to brevity, the notation emphasizes that $P^\perp$ is parabolic, whereas centralizers of arbitrary parabolic subgroups need not be.

We now state two results generalizing results of [17] from nonspherical irreducible parabolic subgroups to essential parabolic subgroups. Lemma 2.6 replaces the conclusions that were previously obtained from the irreducibility hypothesis, and the rest of the proofs go through unchanged.

Lemma 2.8 (cf [17, Lemma 2.48])**.** *Let P be an essential parabolic subgroup of (W,S). Suppose $\Upsilon_0 = w_0\Sigma_{T_0}$ and $\Upsilon_1 = w_1\Sigma_{T_1}$ are elements of $\mathrm{Stab}^{-1}(P)$. Then $T := T_0 = T_1$ and the unique shortest element u of $W_Tw_0^{-1}w_1W_T$ belongs to $W_{T^\perp}$ and satisfies:*

- $\mathfrak{p}_{\Upsilon_1}(\Upsilon_0) = \Upsilon_0^{(0)}u = \Upsilon_1^{(0)}$
- *Left-multiplication by $w_0uw_0^{-1} \in P^\perp$ and right-multiplication by $u \in W_T^\perp$ induce the same isomorphism $\Upsilon_0 \to \Upsilon_1$.*

Lemma 2.9 (cf [17, Corollary 2.17, Lemma 5.3])**.** *Let P be an essential parabolic subgroup of (W,S).*

- *There is a unique parabolic subcomplex $\Upsilon_{N(P)}$ with stabilizer equal to $N_W(P)$.*

- *$P^\perp$ acts freely and transitively by left multiplication on* $\mathrm{Stab}^{-1}(P)$*, and the action induces isomorphisms between the corresponding subcomplexes.*
- *Distinct elements of* $\mathrm{Stab}^{-1}(P)$ *are disjoint, and* $\Upsilon^{(0)}_{N(P)} = \coprod_{\Xi\in\mathrm{Stab}^{-1}(P)} \Xi^{(0)}$.
- *For every* $\Xi \in \mathrm{Stab}^{-1}(P)$*, the composition of the inclusion* $\Xi^{(0)} \hookrightarrow \Upsilon^{(0)}_{N(P)}$ *with the quotient* $\Upsilon^{(0)}_{N(P)} \twoheadrightarrow P^\perp\backslash\Upsilon^{(0)}_{N(P)}$ *is a bijection.*

2.3. **Hyperoctahedrality.** The setup here follows Azuelos and Hagen [4, Section 2.13]. A *crystallographic group* is a group that acts geometrically and faithfully on a Euclidean space $\mathbb{E}^n$. A crystallographic group G has a translation subgroup isomorphic to $\mathbb{Z}^n$, and G modulo its translation group is the finite orthogonal "point group". More generally, for $n \geq 1$, let G be any group with a finite index $\mathbb{Z}^n$ subgroup. By passing to a normal core, G has a finite index normal subgroup $A \cong \mathbb{Z}^n$. Set $V = A \otimes_{\mathbb{Z}} \mathbb{R}$ and let $\rho\colon G \to \mathrm{GL}(V)$ be the representation of G induced by conjugation on A. Its image $F = \rho(G)$ is finite, since $A \leq \ker\rho$ has finite index in G. We again call F the "point group"; in the general case it may be a proper quotient of G/A. When G is crystallographic with translation subgroup T, the representation ρ induces the classical real holonomy representation, which gives an isomorphism $G/T \cong F$.

Definition 2.10 ([4, Definition 2.53], cf [23, Definition 2.2])**.** G is *hyperoctahedral* if V has a basis $\{v_1, \dots, v_n\}$ such that F permutes $\{\pm v_1, \dots, \pm v_n\}$. Equivalently, F is conjugate in $\mathrm{GL}_n(\mathbb{R})$ into the "hyperoctahedral group" of signed permutation matrices $\mathrm{O}_n(\mathbb{Z}) \cong (\mathbb{Z}/2\mathbb{Z})^n \rtimes \mathrm{Sym}_n$.

A different choice of A in G yields a conjugate representation, so ρ is well-defined up to conjugacy, and hyperoctahedrality of G is independent of the choice of A.

The following uses only classical material, but for completeness of Theorem 1.3 we work out the details.

Theorem 2.11 (Hyperoctahedrality classification of irreducible affine Coxeter groups)**.** *The irreducible affine Coxeter groups of types* $\widetilde{A}_1$*,* $\widetilde{A}_3$*,* $\widetilde{B}_n$ *for* $n \geq 2$*,* $\widetilde{C}_n$ *for* $n \geq 2$*, and* $\widetilde{D}_n$ *for* $n \geq 4$ *are hyperoctahedral. The irreducible affine Coxeter groups of types* $\widetilde{A}_2$*,* $\widetilde{A}_n$ *for* $n \geq 4$*,* $\widetilde{E}_6$*,* $\widetilde{E}_7$*,* $\widetilde{E}_8$*,* $\widetilde{F}_4$*, and* $\widetilde{G}_2$ *are nonhyperoctahedral.*

Proof. The irreducible affine Coxeter groups were classified by Coxeter, see [26, Figure 2 in Section 2.5]. Humphreys [26, Chapter 4] constructs a faithful geometric action of the affine Coxeter group of type $\widetilde{X}_n$ on $\mathbb{R}^n$ with point group the finite Weyl group $W(X_n)$ with its standard reflection representation. As a Coxeter group, $W(X_n)$ is the irreducible spherical Coxeter group of type X_n.

Humphreys [26, Section 2.10] works out concrete descriptions of crystallographic root systems. The standard reflection representations for types B_n, C_n, and D_n permute the standard basis of the corresponding $\mathbb{R}^n$, up to sign. Thus, $\widetilde{B}_n$, $\widetilde{C}_n$, and $\widetilde{D}_n$ are hyperoctahedral.

For $\widetilde{A}_n$ the group $W(A_n) \cong \mathrm{Sym}_{n+1}$ acts on $\mathbb{R}^{n+1}$ by permuting its standard basis, but the reflection representation acts on the n–dimensional subspace P of $\mathbb{R}^{n+1}$ consisting of vectors whose coordinates sum to 0. When $n = 1$ the subspace P is spanned by $p_1 = e_1 - e_2$ and the permutation action on $\{e_1, e_2\}$ exchanges p_1 and $-p_1$, so $\widetilde{A}_1$ is hyperoctahedral. For $n = 3$, P has an orthogonal basis $p_1 = (1, 1, -1, -1)$, $p_2 = (1, -1, 1, -1)$, and $p_3 = (1, -1, -1, 1)$, and the set $\{\pm p_1, \pm p_2, \pm p_3\}$ accounts for all of the $\binom{4}{2} = 6$ vectors with two coordinates equal to 1 and two equal to -1, so $W(A_3)$ permutes them. Thus, $\widetilde{A}_3$ is hyperoctahedral.

$W(A_2) \cong \mathrm{Sym}_3$ contains an element of order 3, so it cannot inject into the group $\mathrm{O}_2(\mathbb{Z})$ of order 8.

For $n \geq 4$, suppose $\phi\colon W(A_n) \hookrightarrow \mathrm{O}_n(\mathbb{Z})$. Let K be the kernel of the composition of ϕ with the quotient by the normal $(\mathbb{Z}/2\mathbb{Z})^n$ factor of $\mathrm{O}_n(\mathbb{Z})$. Then K injects via ϕ into $(\mathbb{Z}/2\mathbb{Z})^n$, so it is a normal 2–subgroup of Sym_{n+1}. Since $n + 1 \geq 5$, the only normal subgroups of Sym_{n+1} are 1, Alt_{n+1}, and Sym_{n+1}. The latter two are not 2–groups, so K is trivial. This is impossible, since $|W(A_n)| = |\mathrm{Sym}_{n+1}| > |\mathrm{Sym}_n|$.

For the exceptional types the point group is bigger than the corresponding hyperoctahedral group, see [26, Table 2 in Section 2.11]. □

Example 2.12 ([38] and [4, Figure 1])**.** The group $\widetilde{A}_2 \times \mathbb{Z}$ is hyperoctahedral: its point group is $W(A_2) \cong \mathrm{Sym}_3$ acting by the direct sum of its standard reflection representation and a trivial 1–dimensional representation. As described in the proof of Theorem 2.11, the reflection representation is the coordinate permutation action on the plane $P = \{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid x_1 + x_2 + x_3 = 0\}$. The linear isomorphism $P \oplus \mathbb{R} \cong \mathbb{R}^3 : (p, t) \mapsto p + t(1, 1, 1)$ conjugates the point group action to the coordinate permutation action of Sym_3 on $\mathbb{R}^3$.

Example 2.13. The group $\widetilde{A}_2 \times \widetilde{A}_1$ is not hyperoctahedral. It is virtually $\mathbb{Z}^3$ with point group $F \cong \mathrm{Sym}_3 \times \mathbb{Z}/2\mathbb{Z}$. The conjugation of Example 2.12 takes the reflection of F coming from the $\widetilde{A}_1$ factor to the reflection through P, which is not a signed permutation matrix. This shows why taking an $\widetilde{A}_1$ factor instead of a $\mathbb{Z}$ factor breaks that specific conjugation to signed permutation matrices. To see that no other such conjugation is possible, consider that the 3-dimensional hyperoctahedral group is $\mathrm{O}_3(\mathbb{Z}) \cong (\mathbb{Z}/2\mathbb{Z})^3 \rtimes \mathrm{Sym}_3 \cong \mathrm{Sym}_4 \times \mathbb{Z}/2\mathbb{Z}$. The group F is not conjugate into $\mathrm{O}_3(\mathbb{Z})$ because in F the order 3 rotation coming from the $\widetilde{A}_2$ factor and the reflection coming from the $\widetilde{A}_1$ factor commute. Linear conjugation preserves being a reflection, but there is no pair of commuting order 3 element and reflection in $\mathrm{O}_3(\mathbb{Z})$—one can check every element g of order 3 in $\mathrm{O}_3(\mathbb{Z})$ has centralizer $\langle g \rangle \times \{\pm \mathrm{I}_3\}$, which contains no reflections.

Remark 2.14. Example 2.13 illustrates the difference between F being abstractly isomorphic to a subgroup of $\mathrm{O}_n(\mathbb{Z})$ and being conjugate into $\mathrm{O}_n(\mathbb{Z})$; $\mathrm{O}_3(\mathbb{Z})$ does contain a subgroup isomorphic to $\mathrm{Sym}_3 \times \mathbb{Z}/2\mathbb{Z}$.

2.4. **Hierarchical Hyperbolicity.** Behrstock, Hagen, and Sisto defined hierarchically hyperbolic spaces in [5]. In a sequel [6] the defining axioms were refined. Even for the refined axioms there are two versions: [6, Definition 1.1] and a streamlined version [6, Section 1C], which are equivalent when the Projection Axiom maps are coarsely surjective [6, Proposition 1.11]. The (relative) HHS structure that we will define on a Coxeter group has coarsely surjective projections, so we state the streamlined definition, with the coarse surjectivity hypothesis built in to the Projection Axiom.

Definition 2.15 ([6, Section 1C variant], see also [8, 36])**.** A *hierarchically hyperbolic space* (HHS) consists, for some parameter $E \geq 1$, of an (E, E)–quasigeodesic metric space X, an *index set* $\mathfrak{S}$ of *domains*, and a collection of geodesic, E–hyperbolic *coordinate spaces* $\{(\mathcal{C}P, \delta_P) \mid P \in \mathfrak{S}\}$, satisfying the following axioms:

1. **Projection.** For every domain $P \in \mathfrak{S}$ there is a projection $\pi_P \colon X \to 2^{\mathcal{C}P} \smallsetminus \{\emptyset\}$ such that every point of $\mathcal{C}P$ is within distance E of some point in $\pi_P(X)$ and such that for all $x, y \in X$, $\mathrm{diam}(\pi_P(x) \cup \pi_P(y)) \leq E d_X(x, y) + E$.
2. **Nesting.** If $\mathfrak{S} \neq \emptyset$ then it is partially ordered by the *nesting* relation $\sqsubseteq$, and has a unique maximal element. For $P \in \mathfrak{S}$, define $\mathfrak{S}_P := \{Q \in \mathfrak{S} \mid Q \sqsubseteq P\}$.
3. **Finite complexity.** Chains in the partially ordered set $(\mathfrak{S}, \sqsubseteq)$ have length at most E.
4. **Orthogonality.** $\mathfrak{S}$ has a symmetric, antireflexive *orthogonality* relation $P \perp Q$. Orthogonality is downward closed with respect to nesting and implies incomparability in the nesting ordering:
   (a) $(P \sqsubseteq Q) \wedge (Q \perp R) \implies P \perp R$
   (b) $P \perp Q \implies (P \not\sqsubseteq Q) \wedge (Q \not\sqsubseteq P)$
   For $P \in \mathfrak{S}$, define $\mathfrak{S}_P^{\perp} := \{Q \in \mathfrak{S} \mid P \perp Q\}$.
5. **Containers.** If $P \sqsubsetneq Q$ and $\mathfrak{S}_Q \cap \mathfrak{S}_P^{\perp} \neq \emptyset$ then there exists $R \sqsubsetneq Q$, a *container* of P in Q, such that $\mathfrak{S}_Q \cap \mathfrak{S}_P^{\perp} \subset \mathfrak{S}_R$.
6. **Consistency.** If P and Q are not orthogonal and not nested then they are *transverse*, $P \pitchfork Q$. For all $x \in X$ and $P, Q, R \in \mathfrak{S}$, if $P \sqsubseteq Q$ or $P \pitchfork Q$ there is a specified subset $\emptyset \neq \rho_Q^P \subset \mathcal{C}Q$ of diameter at most E and these subsets satisfy the following consistency conditions:
   (a) $(P \sqsubseteq Q \sqsubsetneq R) \vee (R \not\perp P \sqsubseteq Q \pitchfork R) \implies \delta_R(\rho_R^P, \rho_R^Q) \leq E$
   (b) $P \pitchfork Q \implies \min\{\delta_P(x, \rho_P^Q), \delta_Q(x, \rho_Q^P)\} \leq E$
7. **Bounded geodesic image.** If $P \sqsubsetneq Q$ and there are points $x, y \in X$ with $\delta_P(x, y) > E$ then every geodesic from $\pi_Q(x)$ to $\pi_Q(y)$ in $\mathcal{C}Q$ passes within distance E of ρ_Q^P.
8. **Large links.** For all $x, y \in X$ and $R \in \mathfrak{S}$, there exist domains $P_1, \ldots, P_n$, properly nested in R, with $n \leq N := E\delta_R(x, y) + E$, and such that:
   (a) $\forall i,\ \delta_R(x, \rho_R^{P_i}) \leq N$
   (b) $\forall Q \in \mathfrak{S},\ (Q \sqsubsetneq R) \wedge (\delta_Q(x, y) \geq E) \implies \exists i,\ Q \sqsubseteq P_i$
9. **Partial realization.** For every collection $\{P_i\}$ of pairwise orthogonal domains and points $p_i \in \pi_{P_i}(X) \subset \mathcal{C}P_i$ there exists $x \in X$ such that for all i:
   (a) $\delta_{P_i}(p_i, x) \leq E$
   (b) $\forall Q \in \mathfrak{S},\ (P_i \sqsubsetneq Q) \vee (P_i \pitchfork Q) \implies \delta_Q(x, \rho_Q^{P_i}) \leq E$
10. **Uniqueness.** There is a *uniqueness function* $\theta \colon [0, \infty) \to [0, \infty)$ such that for all $x, y \in X$ and $r \geq 0$, if $d_X(x, y) \geq \theta(r)$ then there exists $P \in \mathfrak{S}$ such that $\delta_P(x, y) \geq r$.

Definition 2.16 (cf [1, Definition 7.1], [36, Definition 3.2])**.** An HHS has *clean containers* if in the Container Axiom it is always possible to choose a container of P in Q that is orthogonal to P.

Definition 2.17 ([6, Definition 6.8],[36, Definition 2.17])**.** A *relatively hierarchically hyperbolic space* is a space X with domains and coordinate spaces satisfying all of the requirements of a hierarchically hyperbolic space of [6, Definition 1.1], except that some of the $\sqsubseteq$–minimal domains are designated *peripheral*, and the coordinate spaces for peripheral domains are not required to be hyperbolic.

Lemma 2.18. *If a space X has a structure $\mathfrak{S}$ satisfying all of the conditions of Definition 2.15, except that some of the $\sqsubseteq$–minimal domains are designated peripheral, and the coordinate spaces of peripheral domains are not required to be hyperbolic, then $\mathfrak{S}$ defines a relative hierarchically hyperbolic structure on X.*

Proof. [6, Proposition 1.11] promotes the streamlined HHS axioms (with uniformly coarsely surjective projections), as enumerated in Definition 2.15, to the full HHS axioms of [6, Definition 1.1]. The step that requires work is to define downward projections $\rho^Q_P\colon \mathcal{C}Q \to 2^{\mathcal{C}P}$ for $P \sqsubsetneq Q \in \mathfrak{S}$. Their construction uses that $\mathcal{C}Q$ is hyperbolic. That is true here, since $P \sqsubsetneq Q$ implies Q is not peripheral. □

Definition 2.19 ([8, Definition 2.27])**.** A finitely generated group G is a *relatively hierarchically hyperbolic group* if it has a Cayley graph X, with respect to some finite generating set, that admits a relatively hierarchically hyperbolic structure $(X, \mathfrak{S})$ such that there is a cofinite action of G on $\mathfrak{S}$ that preserves nesting and orthogonality, preserves the set of peripheral domains, and for which there exists $E \geq 0$ such that for all $g, h \in G$, $P, Q \in \mathfrak{S}$, and $x \in X$, the following conditions are satisfied:

- There is an isometry $g_P\colon \mathcal{C}P \to \mathcal{C}g.P$.
- $(gh)_P = g_{h.P} \circ h_P$
- $\delta_{g.P}(g_P(\pi_P(x)), \pi_{g.P}(gx)) \leq E$
- $(P \sqsubsetneq Q) \vee (P \pitchfork Q) \implies \delta_{g.Q}(g_Q(\rho^P_Q), \rho^{g.P}_{g.Q}) \leq E$

Saying G is *hierarchically hyperbolic relative to* $\mathfrak{P}$ identifies $\mathfrak{P}$ as the set of peripheral domains.

3. Coned off parabolic spaces

The main character of [17] is the space $\widehat{\Sigma}$ obtained from the 1–skeleton of Σ by coning off the vertex set of every wide parabolic subcomplex. It is a hyperbolic, Morse-recognizing space for Σ, by [17, Corollary 4.20].

Lemma 3.1. *Let (W, S) be a nonspherical, finite rank Coxeter system. Suppose W_T is a nontrivial, essential special subgroup of (W, S).*

- *If $N_W(W_T)$ is wide then Σ_T has diameter 1 in $\widehat{\Sigma}$. In particular, this holds when W is wide.*
- *If $N_W(W_T)$ is not wide then Σ_T has infinite diameter in $\widehat{\Sigma}$.*
- *If W is not wide then $\widehat{\Sigma}$ is unbounded hyperbolic.*

Proof. Suppose W_T is a nontrivial, essential special subgroup. Lemma 2.6 says $N_W(W_T) = W_{T \sqcup T^\perp} = W_T \times W_{T^\perp}$. If $N_W(W_T)$ is wide then $\Sigma_{T \sqcup T^\perp}$ is coned off in $\widehat{\Sigma}$, so $\Sigma_T \subset \Sigma_{T \sqcup T^\perp}$ has diameter 1 in $\widehat{\Sigma}$. If W_T is reducible then it is already wide, so $N_W(W_T)$ is too. If W_T is irreducible then Lemma 2.1 says that if W_T is contained in a wide parabolic subgroup then $N_W(W_T)$ is wide. This is the case if W itself is wide.

If $N_W(W_T)$ is not wide then the previous paragraph says that W_T is irreducible and not contained in any wide parabolic subgroup. Take a Coxeter element $g \in W_T$, that is, $g := t_1 \cdots t_n$, where $T = \{t_1, \ldots, t_n\}$. Then g is an infinite order element of W that is not contained in any wide parabolic subgroup, since such a subgroup would also contain $\mathrm{Pc}(g) = W_T$, [35, Theorem 3.1]. [13, Proposition 4.5] says that an infinite order element is rank-one if and only if it is not contained in a wide parabolic subgroup. The Morse-recognizing property of $\widehat{\Sigma}$ then implies that a rank-one element acts loxodromically on $\widehat{\Sigma}$, so $n \mapsto g^n$ is a quasigeodesic. Thus, $\langle g \rangle \subset \Sigma_T$ has infinite diameter in $\widehat{\Sigma}$.

If W is not wide then this argument applies to $W_T = W^{ess}$, so $\widehat{\Sigma}$ is unbounded. □

In [17, Section 5.1] we also defined an auxiliary coned off graph for parabolic subgroups/complexes in which all proper subparabolics are coned off. Given a parabolic subcomplex Υ, define its projection pseudo-metric on $\Sigma^{(0)}$ by $d_\Upsilon(x, y) := d_1(\mathfrak{p}_\Upsilon(x), \mathfrak{p}_\Upsilon(y))$. Define its *coned off graph* to be the metric graph $C(\Upsilon)$ obtained from $\Upsilon^{(1)}$ by coning off the 0-skeleton of every proper parabolic subcomplex Ξ of Υ by adding a new vertex v_Ξ and connecting v_Ξ to every vertex in $\Xi^{(0)}$ by an edge of length $1/2$. We get another projection pseudo-metric on $\Sigma^{(0)}$ by $\delta_\Upsilon(x, y) := d_{C(\Upsilon)}(\mathfrak{p}_\Upsilon(x), \mathfrak{p}_\Upsilon(y))$.

A single nontrivial essential parabolic subgroup P can have many parabolic subcomplexes for which it is the stabilizer. However, Lemma 2.8 and Lemma 2.9 say these P–fibers combine to span the unique parabolic

subcomplex stabilized by $N_W(P) = P \times P^\perp$, and the $P^\perp$–action permutes the P–fibers freely and transitively, inducing isomorphisms between fibers. Furthermore, it follows from Lemma 2.2(3) and (4) that the $P^\perp$–action induces canonical isomorphisms between $C(\Upsilon_0)$ and $C(\Upsilon_1)$ for any $\Upsilon_0, \Upsilon_1 \in \mathrm{Stab}^{-1}(P)$.

This allows us to extend the definitions $\mathfrak{p}_\Upsilon$, d_Υ, $C(\Upsilon)$, and δ_Υ for $\Upsilon \in \mathrm{Stab}^{-1}(P)$ to $\mathfrak{p}_P$, d_P, $C(P)$, and δ_P. Formally, set $\mathfrak{p}_P(x)$ to be the $P^\perp$–orbit of $\mathfrak{p}_\Upsilon(x)$, set $d_P(x,y) := d_1(\mathfrak{p}_P(x), \mathfrak{p}_P(y))$, define $C(P)$ to be the quotient of $\coprod_{\Upsilon' \in \mathrm{Stab}^{-1}(P)} C(\Upsilon')$ by the $P^\perp$–action, and let $\delta_P(x,y) := d_{C(P)}(\mathfrak{p}_P(x), \mathfrak{p}_P(y))$. Then $C(P)$ is canonically isomorphic to $C(\Upsilon)$, and none of these definitions depend on the choice of $\Upsilon \in \mathrm{Stab}^{-1}(P)$.

Lemma 3.2. *There is an ϵ depending on (W,S) such that if P is a nonspherical irreducible parabolic subgroup of (W,S) that is not higher rank affine, then $C(P)$ is ϵ–hyperbolic.*

Moreover, there exists D depending on (W,S) such that for all nonspherical irreducible parabolic subgroups P of (W,S) that are not higher rank affine, all $\Upsilon \in \mathrm{Stab}^{-1}(P)$, all $x, y \in \Upsilon^{(0)}$, all combinatorial geodesics γ from x to y in $\Upsilon^{(1)}$, and all geodesics α from x to y in $C(\Upsilon)$, the Hausdorff distance between α and γ in $C(\Upsilon)$ is at most D.

Proof. Let $\Upsilon \in \mathrm{Stab}^{-1}(P)$. The parabolic subgroup P with its inherited Coxeter system has Davis complex isomorphic to Υ, so, by [17, Corollary 4.20], $\widehat{\Upsilon}$ is an $\epsilon_{\widehat{\Upsilon}}$–hyperbolic graph.

The graph $C(P)$ is canonically isomorphic to the graph $C(\Upsilon)$ obtained from $\Upsilon^{(1)}$ by coning off all proper parabolic subcomplexes. We can perform these cone-offs in two stages, first coning off proper wide parabolic subcomplexes to get $\widehat{\Upsilon}$, and then coning off the rest to get $C(\Upsilon)$. If Ξ is a proper parabolic subcomplex of Υ then Ξ is convex, so every pair of vertices in Ξ is connected by a combinatorial geodesic γ of Σ that stays inside of Ξ. In $\widehat{\Upsilon}$, the image of γ is an unparameterized uniform quasigeodesic [17, Corollary 4.5 and Corollary 4.20], which is therefore uniformly Morse, since $\widehat{\Upsilon}$ is hyperbolic. It follows that the image of Ξ in $\widehat{\Upsilon}$ is quasiconvex. Thus, $C(\Upsilon)$ is obtained from the hyperbolic graph $\widehat{\Upsilon}$ by coning off a collection of uniformly quasiconvex subspaces. A result of Kapovich and Rafi [27, Proposition 2.6] then implies that $C(\Upsilon)$ is ϵ_Υ–hyperbolic, for some ϵ_Υ depending on $\epsilon_{\widehat{\Upsilon}}$ and the quasiconvexity constant. Set $\epsilon_P := \epsilon_\Upsilon$, which is independent of the choice of $\Upsilon \in \mathrm{Stab}^{-1}(P)$. Moreover, conjugates of P have fibers that differ by isometries of Σ, so ϵ_P only depends on the conjugacy class of P. Since there are only finitely many conjugacy classes of parabolic subgroup of (W,S), we get uniform ϵ by maximizing over the finitely many values ϵ_P.

For the Hausdorff claim, γ becomes a uniform unparameterized quasigeodesic in the hyperbolic space $\widehat{\Upsilon}$, so it is uniformly Hausdorff close to any $\widehat{\Upsilon}$–geodesic β with the same endpoints. The map from $\widehat{\Upsilon}$ to $C(\Upsilon)$ is 1–Lipschitz, so γ and β remain uniformly Hausdorff close in $C(\Upsilon)$. The second part of [27, Proposition 2.6] says that β is uniformly Hausdorff close to any $C(\Upsilon)$–geodesic α with the same endpoints. This gives an Υ–specific Hausdorff distance bound, but again it actually is conjugacy invariant, so we can maximize over the finitely many conjugacy classes of parabolics to get a uniform bound. □

Proposition 3.3. *Let P be a nontrivial essential parabolic subgroup.*

- *If P is reducible then $C(P)$ has diameter at most 3.*
- *If P is irreducible affine then $C(P)$ is quasiisometric to $\mathbb{Z}^{\mathrm{rank}(P)-1}$.*
- *If P is irreducible and not higher rank affine then $C(P)$ is hyperbolic and unbounded.*

The second and third cases overlap for infinite dihedral P.

Proof. It suffices to assume $P = W_T$ is a special subgroup. If it is essential and reducible then let W_{T_0} be one nonspherical irreducible factor of W_T and let $T_1 := T \smallsetminus T_0$. For all $x, y \in W_T$, write $x^{-1}y = ab$ for $a \in W_{T_0}$ and $b \in W_{T_1}$. Then x and xa are vertices contained in the proper parabolic subcomplex $x\Sigma_{T_0}$ of Σ_T, while xa and $y = xab$ are contained in the proper parabolic subcomplex $xa\Sigma_{T_1}$ of Σ_T. Both of these proper parabolic subcomplexes are coned off in $C(\Sigma_T)$, so $d_{C(\Sigma_T)}(x,y) \leq 2$. Every cone vertex of $C(\Sigma_T)$ is at distance $1/2$ from some $\Sigma_T^{(0)}$ vertex, so 3 is an upper diameter bound for all vertices of $C(\Sigma_T)$.

Suppose $P = W_T$ is irreducible and affine. Every proper parabolic subgroup of an irreducible affine Coxeter group is spherical, so $C(\Sigma_T)$ is constructed from the 1–skeleton of Σ_T by coning off vertices of spherical parabolic subcomplexes. Spherical parabolic subcomplexes of Σ, hence of Σ_T, have uniformly bounded diameter, so $C(\Sigma_T)$ is quasiisometric to Σ_T, hence is unbounded, since W_T is nonspherical. Since W_T is affine, it has a finite index translation subgroup isomorphic to $\mathbb{Z}^{|T|-1}$. Thus, $C(\Sigma_T) \cong C(W_T)$, Σ_T, W_T, and $\mathbb{Z}^{|T|-1}$ are all quasiisometric to one another.

Now suppose $P = W_T$ is irreducible and not higher rank affine. Let $T = \{t_0, \dots, t_{r-1}\}$. Take a Coxeter element $c := t_0 \cdots t_{r-1}$ of W_T, which has $\mathrm{Pc}(c) = W_T$, [35, Theorem 3.1].

By [17, Corollary 4.20], $\widehat{\Sigma}_T$ is hyperbolic, and the proof of Lemma 3.1 shows that c acts as a rank-one element on Σ_T and acts loxodromically on $\widehat{\Sigma}_T$. The proof of Lemma 3.2 shows that $C(\Sigma_T)$ is a hyperbolic graph obtained from $\widehat{\Sigma}_T$ by coning off further proper parabolic subcomplexes of Σ_T. It remains to show that $C(\Sigma_T)$ has unbounded diameter, which we establish by showing that c remains loxodromic in $C(\Sigma_T)$.

Let γ be the biinfinite path through 1 in Σ_T that starting from 1 in one direction has edge labels t_0, t_1, $\dots$,t_{r-1}, t_0,$\dots$, and in the other direction has edge labels t_{r-1}, t_{r-2}, $\dots$, t_0, t_{r-1},$\dots$, so that γ is an axis for the c–action. By [39, Theorem 1], γ is a combinatorial geodesic.

Since c acts as a rank-one element on Σ_T, its axis γ is a Morse quasigeodesic in the CAT(0) metric, hence is a Morse geodesic in the combinatorial metric. Thus, there exists R such that every combinatorial geodesic segment with endpoints on γ is contained in the R–neighborhood of γ, in the combinatorial metric d_1.

For $n > 0$ suppose $m := d_{C(\Sigma_T)}(1, c^n)$. By [17, Lemma 5.21], there is a factorization $c^n = h_1 \cdots h_m$ such that for each i there is $J_i \subsetneq T$ with $h_i \in W_{J_i}$ and a geodesic β from 1 to c^n that is a concatenation of geodesic subsegments β_i labelled by h_i.

Let a_i and b_i be the end vertices of β_i. There exist a_i' and b_i' on γ at distance at most R from a_i and b_i, respectively. We can choose $a_1' = a_1$, $a_{i+1}' = b_i'$, and $b_m' = b_m$.

Suppose, for some i, that $d_1(a_i', b_i') \geq r(2R+1)$. Subdivide the subsegment γ_i of γ between a_i' and b_i' into (possibly trivial) subsegments such that each edge is dual to a wall that cuts the proper parabolic subcomplex Υ_i of type J_i containing β_i, separated by single edges dual to a wall that does not cut Υ_i. Now, a wall M that separates a_i' and b_i' but does not cut Υ_i does not separate a_i and b_i, so either M separates a_i and a_i' or M separates b_i and b_i'. There are at most R walls in each of those cases, so γ_i is subdivided into at most $2R+1$ subsegments with all edges cut by a wall that cuts Υ_i, separated by $2R$ single edges that do not cut Υ_i. By the distance hypotheses, at least one of the former has length at least r, so γ_i contains a subsegment of length r whose edges are dual to walls M_1, $\dots$, M_r with each M_i cutting Υ_i. On one hand, that means that the reflections in the walls M_j are all in $\mathrm{Stab}(\Upsilon_i) \subsetneq W_T$. On the other hand, if g is the initial vertex of the edge cut by M_1, then there is $0 \leq k < r$ such that, taking all subscripts modulo r, the reflection in M_1 is gt_kg^{-1}, the reflection in M_2 is $gt_kt_{k+1}t_kg^{-1}$,$\dots$, the reflection in M_r is $gt_k \cdots t_{k+r-2}t_{k+r-1}t_{k+r-2} \cdots t_kg^{-1}$. Thus the subgroup generated by the reflections in the walls M_1,$\dots$,M_r is $g\langle t_0, \dots, t_{r-1}\rangle g^{-1} = W_T$. This is a contradiction, so $d_1(a_i', b_i') < r(2R+1)$. Since this bound holds for each i:

$$nr = |c^n|_T < m \cdot r(2R+1) \implies d_{C(\Sigma_T)}(1, c^n) = m > \frac{n}{2R+1}$$

Thus, c acts loxodromically on $C(\Sigma_T)$, which implies that $C(\Sigma_T) \cong C(W_T)$ is unbounded. □

4. Relative HHG structure on (W, S)

To prove Theorem 1.1, we define a relative HHS structure on $\Sigma^{(1)} = \mathrm{Cay}(W, S)$ in Section 4.1, verify the HHS axioms in Section 4.2, and then check the relative HHG equivariance conditions in Section 4.3.

4.1. Defining the relative HHS structure.

Given a finite rank Coxeter system (W, S), define the index set $\mathfrak{S}$ of domains to consist of:

(1) A single, formal, top domain W.
(2) Every nontrivial, essential parabolic subgroup P of (W, S) such that $P \neq W^{ess}$ and $N_W(P)$ is wide.

Define $\mathsf{W} \sqsubseteq \mathsf{W}$ and $\mathsf{W} \not\perp \mathsf{W}$, and for all $P \in \mathfrak{S} \smallsetminus \{\mathsf{W}\}$, define $\mathsf{W} \not\sqsubseteq P \sqsubseteq \mathsf{W}$ and $P \not\perp \mathsf{W}$. For $P, Q \in \mathfrak{S} \smallsetminus \{\mathsf{W}\}$, define $P \sqsubseteq Q$ if and only if $P \leq Q$, and define $P \perp Q$ if and only if P and Q commute (element-wise).

By Moussong's Theorem [31], W is hyperbolic if and only if it has no wide parabolic subgroups. In this case, $\mathfrak{S} = \{\mathsf{W}\}$ and W is already an HHG by taking $\mathcal{C}\mathsf{W} = \Sigma^{(1)}$. If W^{ess} is irreducible higher rank affine then it has no nonspherical proper parabolic subgroups, so again $\mathfrak{S} = \{\mathsf{W}\}$. Set $\mathcal{C}\mathsf{W} := \Sigma^{(1)}$. This is a trivial HHG structure on W relative to $\mathfrak{S}$. In both of these cases, Theorem 1.1 is confirmed, so assume we are not in either. It follows that $\mathfrak{S} \neq \{\mathsf{W}\}$.

If W is wide, equivalently, since we have assumed that W^{ess} is not higher rank affine, if W^{ess} is reducible, define $\mathcal{C}\mathsf{W}$ to be a point, and define π_{W} to be the unique map $\Sigma^{(0)} \to \mathcal{C}\mathsf{W}$.

If W is not wide then define $\mathcal{C}\mathsf{W} := \widehat{\Sigma}$, and define π_{W} to be the canonical map identifying $\Sigma^{(0)}$ with the non-cone vertices of $\widehat{\Sigma}$.

For reducible $P \in \mathfrak{S} \smallsetminus \{\mathsf{W}\}$, define $\mathcal{C}P$ to be a single point, and define π_P to be the unique map $\Sigma^{(0)} \to \mathcal{C}P$.

For irreducible $P \in \mathfrak{S} \smallsetminus \{\mathsf{W}\}$, define $\mathcal{C}P := C(P)$ and $\pi_P := \mathfrak{p}_P$, as in Section 3.

The formal top domain W is in many ways playing the role of W^{ess}, and so we will say W is ir/reducible if W^{ess} is ir/reducible. We formally distinguish W from W^{ess} because:

(1) We do not require that $N_W(W^{ess}) = W$ is wide.
(2) In the irreducible case, $\mathcal{C}\mathsf{W}$ is $\widehat{\Sigma}$, not $C(W^{ess})$.

We chose to make the coordinate spaces for reducible domains single points, rather than the bounded spaces $\widehat{\Sigma}$ or $C(P)$ that would have resulted from a uniform definition. This emphasizes that they are present in the hierarchy for formal reasons (to provide a maximal element and to provide containers).

Theorem 1.1 says the peripherals are the irreducible higher rank affine parabolic subgroups. By construction, these are all domains. We now check that they are precisely the domains with non-hyperbolic coordinate spaces, and that they are all $\sqsubseteq$–minimal:

Lemma 4.1. *A domain $P \in \mathfrak{S}$ has non-hyperbolic coordinate space $\mathcal{C}P$ if and only if P is an irreducible higher rank affine parabolic subgroup, and every such domain is $\sqsubseteq$–minimal. The other domains have uniformly hyperbolic coordinate spaces.*

Proof. For reducible $P \in \mathfrak{S}$ the coordinate space $\mathcal{C}P$ is a single point, hence 0–hyperbolic.

If W is irreducible then $\mathcal{C}\mathsf{W} = \widehat{\Sigma}$ is unbounded hyperbolic, by Lemma 3.1.

If P is an irreducible affine parabolic subgroup then, as in the proof of Proposition 3.3, P has no proper nonspherical parabolic subgroups and is quasiisometric to $\mathbb{Z}^{\mathrm{rank}(P)-1}$. The former implies P is $\sqsubseteq$–minimal in $\mathfrak{S}$, and the latter implies $\mathcal{C}P$ is non-hyperbolic when $\mathrm{rank}(P) \geq 3$.

For non higher rank affine, irreducible $P \in \mathfrak{S} \smallsetminus \{\mathsf{W}\}$, Proposition 3.3 says $\mathcal{C}P$ is unbounded hyperbolic. □

4.2. **Verifying the HHS axioms.** We verify the HHS axioms one at a time.

4.2.1. *Projections.* All the projections are 1–Lipschitz and 1–coarsely surjective.

4.2.2. *Nesting.* The parabolic subgroups of (W, S) are partially ordered by inclusion. Our $(\mathfrak{S}, \sqsubseteq)$ is a partially ordered subset of that partially ordered set, to which we added a formal maximal element W.

4.2.3. *Finite complexity.* If $P \lneq Q$ are nontrivial essential parabolic subgroups then $2 \leq \mathrm{rank}(P) < \mathrm{rank}(Q)$. Since non-top domains are proper nontrivial essential parabolic subgroups properly contained in W^{ess}, their ranks are between 2 and $\mathrm{rank}(W^{ess}) - 1$, so a chain of them has length at most $\mathrm{rank}(W^{ess}) - 2$. Adding W gives that chains in $(\mathfrak{S}, \sqsubseteq)$ have length at most $\mathrm{rank}(W^{ess}) - 1 \leq |S| - 1$.

4.2.4. *Orthogonality.* By definition, for all $P \in \mathfrak{S}$, we have $P \sqsubseteq \mathsf{W}$ and $P \not\perp \mathsf{W}$. For $P, Q \in \mathfrak{S} \smallsetminus \{\mathsf{W}\}$, if $P \sqsubseteq Q$ then $P \leq Q$. Let $P = P_1 \times \cdots \times P_n$ be the canonical decomposition of P into nonspherical irreducible factors. If P and Q commute then, in particular, P_1 commutes with itself, but that is impossible, since nonspherical irreducible parabolic subgroups have trivial center. Thus, $P \sqsubseteq Q$ implies $P \not\perp Q$.

We have shown $P \sqsubseteq Q$ implies $P \not\perp Q$, so, contrapositively, if $P \perp Q$ then $P \not\sqsubseteq Q$ and $Q \not\sqsubseteq P$.

Now suppose $P, Q, R \in \mathfrak{S}$ with $P \sqsubseteq Q$ and $Q \perp R$. No domain is orthogonal to W, so $P, Q, R \in \mathfrak{S} \smallsetminus \{\mathsf{W}\}$ are parabolic subgroups, and we have that $P \leq Q$ and Q and R commute, so P and R commute, so $P \perp R$.

4.2.5. *Clean containers.* Consider $P, Q \in \mathfrak{S} \smallsetminus \{\mathsf{W}\}$, with $P \sqsubsetneq Q$. If $\mathfrak{S}_Q \cap \mathfrak{S}_P^{\perp} = \emptyset$ there is nothing to prove, so suppose $R \in \mathfrak{S}_Q \cap \mathfrak{S}_P^{\perp}$. We claim that $(Q \cap P^{\perp})^{ess}$ is a clean container for P in Q. It is a proper essential parabolic subgroup that commutes with P and is contained in Q. It is not equal to Q, since $P \sqsubseteq Q$ implies P and Q do not commute. Furthermore, R is contained in Q and commutes with P, so $R \leq Q \cap P^{\perp}$, so, by Lemma 2.5, $R = R^{ess} \leq (Q \cap P^{\perp})^{ess}$. It remains only to show that $(Q \cap P^{\perp})^{ess}$ is actually a domain. It is nontrivial, since it contains R, and is a proper essential parabolic subgroup by construction. To conclude it is a domain, observe, by Lemma 2.6, that $N_W((Q \cap P^{\perp})^{ess}) = (Q \cap P^{\perp})^{ess} \times Z_W((Q \cap P^{\perp})^{ess})$, which is wide, because the first factor contains R and the second factor contains P, so both factors are nonspherical.

A similar argument shows that $(P^{\perp})^{ess} = (W^{ess} \cap P^{\perp})^{ess}$ is a clean container for P in W.

4.2.6. *Consistency.*

Definition 4.2. If P is a nontrivial, proper, essential parabolic subgroup of (W,S), recall that $\Upsilon_{N(P)}$ denotes the unique parabolic subcomplex of Σ with stabilizer equal to $N_W(P)$.

For $P \in \mathfrak{S} \smallsetminus \{\mathsf{W}\}$ and $Q \in \mathfrak{S}$, define $\rho^P_Q = \pi_Q(\Upsilon_{N(P)})$.

When Q is reducible the sets ρ^P_Q have diameter 0. When W is irreducible, the fact that ρ^P_{W} has diameter 1 in $\mathcal{C}\mathsf{W}$ for all $P \subsetneqq \mathsf{W}$ is the easy direction of Lemma 3.1. The following handles the remaining case:

Lemma 4.3. *For $P \pitchfork Q$ or $P \subsetneqq Q \subsetneqq \mathsf{W}$ with irreducible Q, the set ρ^P_Q has diameter at most 1 in $\mathcal{C}Q$.*

Proof. First observe:

$$(P \pitchfork Q) \vee (P \subsetneqq Q) \implies (P \not\perp Q) \wedge (Q \not\sqsubseteq P)$$

Pick $\Xi \in \mathrm{Stab}^{-1}(Q)$, and let $\Upsilon_{N(P)}$ be the unique element of $\mathrm{Stab}^{-1}(N_W(P))$. The canonical isomorphism between $\mathcal{C}Q$ and $C(\Xi)$ takes ρ^P_Q to $\mathfrak{p}_\Xi(\Upsilon_{N(P)})$. If $\mathfrak{p}_\Xi(\Upsilon_{N(P)})$ is all of $\Xi^{(0)}$ then, using Lemma 2.2(6):

$$Q = \mathrm{Stab}(\Xi) = \mathrm{Stab}(\mathfrak{p}_\Xi(\Upsilon_{N(P)})) = Q \cap N_W(P) \leq N_W(P) = P \times P^\perp$$

Since Q is irreducible, $Q \leq P \times P^\perp$ implies that either $Q \leq P$, which contradicts $Q \not\sqsubseteq P$, or $Q \leq P^\perp$, which contradicts $P \not\perp Q$. Since both outcomes are contradictions, $\mathfrak{p}_\Xi(\Upsilon_{N(P)})$ is not all of $\Xi^{(0)}$. Rather, by Lemma 2.2(5), it spans a proper parabolic subcomplex of Ξ, so it is coned off in $C(\Xi)$. □

The consistency condition 6(a) is immediate: $(P \sqsubseteq Q \subsetneqq R) \vee (R \not\perp P \sqsubseteq Q \pitchfork R)$ implies ρ^P_R and ρ^Q_R are both defined, and that $P \leq Q$ are parabolic subgroups. Choose $\Xi \in \mathrm{Stab}^{-1}(Q)$. There exists $\Upsilon \in \mathrm{Stab}^{-1}(P)$ such that $\Upsilon \subset \Xi$. Thus, the image of Υ is contained in $\rho^P_R \cap \rho^Q_R$, so $\delta_R(\rho^P_R, \rho^Q_R) = 0$.

The consistency condition 6(b) is given by the following, whose proof is a "grid of grids" argument inspired by the proof of [14, Theorem 2.16]. Recall that the wall structure of the Niblo-Reeves cube complex $X_{\mathbb{NR}}$ of (W,S) is W–equivariantly isomorphic to that of Σ. Since $X_{\mathbb{NR}}$ is a CAT(0) cube complex, its walls have the Helly property. It follows that the maximum size of a set of pairwise transverse walls of Σ is $\dim(X_{\mathbb{NR}})$.

Proposition 4.4. *For all $x \in \Sigma^{(0)}$ and $P, Q \in \mathfrak{S}$ with $P \pitchfork Q$:*

$$\min\{\delta_P(x, \rho^Q_P), \delta_Q(x, \rho^P_Q)\} \leq 1 + 2(2\ell - 1)\dim(X_{\mathbb{NR}})$$

Proof. If either of P or Q is reducible then the corresponding coordinate space is a single point, so the minimum in the statement is 0. Thus, we may assume that both P and Q are irreducible.

Suppose the proposition is not satisfied, so both domain distances are strictly greater than the given bound. Choose $\Upsilon \in \mathrm{Stab}^{-1}(P)$ and $\Xi \in \mathrm{Stab}^{-1}(Q)$. Define $a := \mathfrak{p}_\Upsilon(x)$, $b := \mathfrak{p}_{\mathfrak{p}_\Upsilon(\Xi)}(x)$, $c := \mathfrak{p}_\Xi(x)$, and $d := \mathfrak{p}_{\mathfrak{p}_\Xi(\Upsilon)}(x)$. Then $d_{C(\Upsilon)}(a,b) \geq d_{C(\Upsilon)}(a, \mathfrak{p}_\Upsilon(\Xi)) \geq \delta_P(x, \Xi) \geq \delta_P(x, \rho^Q_P)$, and, similarly, $d_{C(\Xi)}(c,d) \geq \delta_Q(x, \rho^P_Q)$.

Every wall $V \in \mathcal{M}(a,b)$ separating a from b cuts Υ, by convexity of Υ. Furthermore, we claim that V separates x from Ξ. To see this, suppose V does not separate x and some $z \in \Xi$. Observe, by Lemma 2.2(1), that $b = \mathfrak{p}_{\mathfrak{p}_\Upsilon(\Xi)}(a)$, so by Lemma 2.2(2), V separates $a = \mathfrak{p}_\Upsilon(x)$ from $\mathfrak{p}_\Upsilon(\Xi)$. Since $\mathfrak{p}_\Upsilon(\Xi)$ contains $\mathfrak{p}_\Upsilon(z)$, either V separates z from $\mathfrak{p}_\Upsilon(z)$ or V separates x from $\mathfrak{p}_\Upsilon(x)$. By Lemma 2.2(2), this implies that V separates z from Υ or V separates x from Υ. Both are impossible, since V cuts Υ.

Similarly, every wall $H \in \mathcal{M}(c,d)$ cuts Ξ and separates x from Υ.

If we label the complementary open halfspaces so that $x \in V^+ \cap H^+$, then we have $a \in V^+ \cap H^-$, $b, d \in V^- \cap H^-$, and $c \in V^- \cap H^+$. Since all four halfspace intersections are nonempty, $V \pitchfork H$. Since V and H were arbitrary, every wall of $\mathcal{M}(a,b)$ is transverse to every wall of $\mathcal{M}(c,d)$.

Each wall of $\mathcal{M}(a,b)$ cuts Υ, so $\mathrm{Pc}(\mathcal{M}(a,b)) \leq P$. If it were a proper parabolic subgroup of P then a and b would be contained in a proper parabolic subcomplex of Υ. That is impossible, because then $d_{C(\Upsilon)}(a,b) \leq 1$, contrary to hypothesis. Thus $\mathrm{Pc}(\mathcal{M}(a,b)) = P$. In fact, we can throw away some of the walls without the parabolic closure getting smaller: suppose $\mathcal{M}'$ is a subset of $\mathcal{M}(a,b)$ for which $\mathrm{Pc}(\mathcal{M}(a,b) \smallsetminus \mathcal{M}') \subsetneq P$. Take a combinatorial geodesic γ from a to b, which is contained in Υ, by convexity. The edges of γ are in bijection with $\mathcal{M}(a,b)$, so $\mathcal{M}'$ gives a subdivision of γ into a concatenation of (possibly trivial) subsegments with all edges dual to a wall of $\mathcal{M}(a,b) \smallsetminus \mathcal{M}'$, separated by single edges dual to a wall of $\mathcal{M}'$. By hypothesis, each of the former type of subpath has every reflection of its dual walls in $\mathrm{Pc}(\mathcal{M}(a,b) \smallsetminus \mathcal{M}') \subsetneq P$. Thus, its endpoints are contained in a proper parabolic subcomplex of Υ, so they are at distance at most 1 in $C(\Upsilon)$. This implies that:

$$1 + 2(2\ell - 1)\dim(X_{\mathbb{NR}}) < d_{C(\Upsilon)}(a,b) \leq 2|\mathcal{M}'| + 1$$

Hence, $|\mathcal{M}'| > (2\ell-1)\dim(X_{\text{NR}})$. Conversely:

$$|\mathcal{M}'| \leq (2\ell-1)\dim(X_{\text{NR}}) \implies \text{Pc}(\mathcal{M}(a,b)\smallsetminus\mathcal{M}') = P \tag{1}$$

Consider the set of complementary open halfspaces of walls of $\mathcal{M}(a,b)$ containing a, partially ordered by inclusion. By Dilworth's Theorem, that partially ordered set can be partitioned into a collection of disjoint chains, the number of which is at most the maximal size of an antichain in the collection. In this setup, chains of nested halfspaces correspond to chains of walls of $\mathcal{M}(a,b)$, and maximal antichains come from maximal subsets of $\mathcal{M}(a,b)$ of pairwise crossing walls. The cardinality of any set of pairwise crossing walls of Σ is at most $\dim(X_{\text{NR}})$. Conclude that $\mathcal{M}(a,b)$ can be partitioned into at most $\dim(X_{\text{NR}})$ disjoint chains.

Take such a partition of $\mathcal{M}(a,b)$ into at most $\dim(X_{\text{NR}})$ disjoint chains, and throw away any whose length is at most $2\ell-1$. Label the remaining chains $\mathcal{V}_1,\dots,\mathcal{V}_m$. Consider the set of walls $\mathcal{M}''$ in the interior of one of the $\mathcal{V}_i$, and take $\mathcal{M}' := \mathcal{M}(a,b)\smallsetminus\mathcal{M}''$. Then the total number $|\mathcal{M}'|$ of walls discarded, either for belonging to a short chain or for being near an end of a long chain, is:

$$|\mathcal{M}'| \leq (\dim(X_{\text{NR}})-m)\cdot(2\ell-1)+m(2\ell-2) \leq (2\ell-1)\dim(X_{\text{NR}})$$

By (1), $\text{Pc}(\mathcal{M}'') = \text{Pc}(\mathring{\mathcal{V}}_1,\dots,\mathring{\mathcal{V}}_m) = P$. This gives $P = \text{Pc}(\mathring{\mathcal{V}}_1,\dots,\mathring{\mathcal{V}}_m) \leq \text{Pc}(\mathcal{V}_1,\dots,\mathcal{V}_m) \leq P$.

By a symmetric argument applied to $\mathcal{M}(c,d)$, we get disjoint chains $\mathcal{H}_1,\dots,\mathcal{H}_n$ in $\mathcal{M}(c,d)$ that each have length at least 2ℓ and such that $\text{Pc}(\mathcal{H}_1,\dots,\mathcal{H}_n) = \text{Pc}(\mathring{\mathcal{H}}_1,\dots,\mathring{\mathcal{H}}_n) = Q$.

For each i,j, $(\mathcal{V}_i,\mathcal{H}_j)$ is a grid with $|\mathcal{V}_i|,|\mathcal{H}_j| \geq 2\ell$, so by [17, Proposition 3.20], either $\text{Pc}(\mathcal{V}_i) = \text{Pc}(\mathcal{H}_j)$ is irreducible higher rank affine, or $\text{Pc}(\mathring{\mathcal{H}}_j) \leq \text{Pc}(\mathcal{V}_i)^\perp$.

First suppose the latter outcome holds for every pair of indices i,j. Then for all i, for all j, $\text{Pc}(\mathring{\mathcal{H}}_j) \leq \text{Pc}(\mathcal{V}_i)^\perp$, and since $\text{Pc}(\mathcal{V}_i)^\perp$ is a parabolic subgroup, $Q = \text{Pc}(\mathring{\mathcal{H}}_1,\dots,\mathring{\mathcal{H}}_n) \leq \text{Pc}(\mathcal{V}_i)^\perp$. Thus, for all i, $\text{Pc}(\mathcal{V}_i) \leq Q^\perp$. But $Q^\perp$ is a parabolic subgroup, so $P = \text{Pc}(\mathcal{V}_1,\dots,\mathcal{V}_m) \leq Q^\perp$. This says P and Q commute, which is a contradiction, because that means $P \perp Q$, but we started with $P \pitchfork Q$.

The other option is that there is some pair of indices, which, up to relabelling, we may assume are both 1, such that $\text{Pc}(\mathcal{V}_1) = \text{Pc}(\mathcal{H}_1)$ is irreducible higher rank affine. For all $1 < i \leq m$, consider the grid $(\mathcal{H}_1,\mathcal{V}_i)$. By [17, Proposition 3.20], either $\text{Pc}(\mathring{\mathcal{V}}_i) \leq \text{Pc}(\mathcal{H}_1)^\perp$, or $\text{Pc}(\mathring{\mathcal{V}}_i) \leq \text{Pc}(\mathcal{V}_i) = \text{Pc}(\mathcal{H}_1)$. Then $\text{Pc}(\mathcal{H}_1)\times\text{Pc}(\mathcal{H}_1)^\perp$ is a parabolic subgroup that contains each $\text{Pc}(\mathring{\mathcal{V}}_i)$, so contains the irreducible parabolic subgroup $P = \text{Pc}(\mathring{\mathcal{V}}_1,\dots,\mathring{\mathcal{V}}_m)$. Since P is irreducible it is contained in either $\text{Pc}(\mathcal{H}_1)$ or $\text{Pc}(\mathcal{H}_1)^\perp$, but since we already know that P has nontrivial intersection with $\text{Pc}(\mathcal{H}_1)$, as $\text{Pc}(\mathcal{H}_1) = \text{Pc}(\mathcal{V}_1) \leq P$, it follows that $P = \text{Pc}(\mathcal{H}_1) \leq Q$, which implies $P \sqsubseteq Q$, which contradicts $P \pitchfork Q$. □

4.2.7. *Bounded geodesic image.* If $P \not\sqsubseteq Q$ and P is reducible then $\mathcal{C}P$ is a point, so $\delta_P(x,y) > E$ never occurs. If Q is reducible then $\mathcal{C}Q$ is a point, so the conclusion is immediate. If P is irreducible and $Q = \bowtie$ is irreducible then, up to taking large enough E, this is [17, Corollary 5.25]. Thus, the remaining case is that both P and Q are irreducible with $Q \neq \bowtie$:

Proposition 4.5. *There exist B and K such that if $P \lneq Q$ are pre-wide parabolic subgroups, $x,y \in \Sigma^{(0)}$, and γ is a geodesic from $\pi_Q(x)$ to $\pi_Q(y)$ in $\mathcal{C}Q$, then:*

$$\delta_Q(\gamma,\rho_Q^P) > B \implies \delta_P(x,y) \leq K$$

Proof. Note that Q is not higher rank affine, since it contains the proper nonspherical subparabolic P, so $\mathcal{C}Q$ is ϵ–hyperbolic, by Lemma 4.1. Let D be the Hausdorff distance bound of Lemma 3.2. We will show it suffices to take $K := 2D+2\epsilon$ and $B := C+2\epsilon+1$ for $C := 2D+2\epsilon+2$.

Choose $\Xi \in \text{Stab}^{-1}(Q)$. The coordinate space $\mathcal{C}Q$ is canonically isomorphic to $C(\Xi)$, by a map that identifies $\pi_Q(x)$ to $\mathfrak{p}_\Xi(x)$ and ρ_Q^P with $\rho_\Xi^P := \bigcup_{\Upsilon\in\text{Stab}^{-1}(P)}\mathfrak{p}_\Xi(\Upsilon)$. Suppose that $x,y \in \Sigma^{(0)}$ are vertices and γ is a geodesic from $x' := \mathfrak{p}_\Xi(x)$ to $y' := \mathfrak{p}_\Xi(y)$ in $C(\Xi)$ such that $d_{C(\Xi)}(\gamma,\rho_\Xi^P) > B$. Choose $\Upsilon \in \text{Stab}^{-1}(P)$ with $\Upsilon \subset \Xi$, and let $x'' := \mathfrak{p}_\Upsilon(x) = \mathfrak{p}_\Upsilon(x')$ and $y'' := \mathfrak{p}_\Upsilon(y) = \mathfrak{p}_\Upsilon(y')$, where the equalities are from Lemma 2.2(1).

Let β_x be a combinatorial geodesic in $\Sigma^{(1)}$ from x' to x'', and let β_y be combinatorial geodesic from y' to y''. Let $\bar\beta_x$ be a $C(\Xi)$–geodesic from x' to x'', and let $\bar\beta_y$ be a $C(\Xi)$–geodesic from y' to y''. Let α be a $C(\Xi)$–geodesic from x'' to y'', so that α, $\bar\beta_x$, $\bar\beta_y$, and γ are the sides of a geodesic quadrilateral in the ϵ–hyperbolic space $C(\Xi)$. That quadrilateral is 2ϵ–thin.

Observe that $\Upsilon = \mathfrak{p}_\Xi(\Upsilon) \subset \rho_\Xi^P$, so $d_{C(\Xi)}(x',x'') \geq d_{C(\Xi)}(\gamma,\rho_\Xi^P) > B$ so there is a point $\bar b_x \in \bar\beta_x$ at distance C from x''. We claim that $\bar b_x$ is more than 2ϵ from α and γ, so it is 2ϵ–close to some point $\bar b_y \in \bar\beta_y$. First, since

$\operatorname{diam}_{d_{C(\Xi)}}(\rho^P_\Xi) \leq 1$ and since α connects points x'' and y'' of ρ^P_Ξ, the length of α is at most 1, so $d_{C(\Xi)}(\bar{b}_x, \alpha) \geq d_{C(\Xi)}(\bar{b}_x, x'') - 1 = C - 1 > 2\epsilon$. Second, $d_{C(\Xi)}(\bar{b}_x, \gamma) \geq d_{C(\Xi)}(x'', \gamma) - d_{C(\Xi)}(x'', \bar{b}_x) > B - C > 2\epsilon$.

By Lemma 3.2, there exist $b_x \in \beta_x$ and $b_y \in \beta_y$ such that $d_{C(\Xi)}(\bar{b}_x, b_x) \leq D$ and $d_{C(\Xi)}(\bar{b}_y, b_y) \leq D$. Observe, $d_{C(\Xi)}(b_x, b_y) \leq 2\epsilon + 2D = K$.

Also, $d_{C(\Xi)}(b_x, \rho^P_\Xi) \geq d_{C(\Xi)}(\bar{b}_x, \rho^P_\Xi) - D \geq d_{C(\Xi)}(\bar{b}_x, x'') - 1 - D \geq C - 1 - D$, and, similarly, $d_{C(\Xi)}(b_y, \rho^P_\Xi) \geq C - 1 - D - 2\epsilon$. Let η be a $C(\Xi)$–geodesic from b_x to b_y. It has length at most K, and for all $c \in \eta$:

$$\begin{aligned} d_{C(\Xi)}(c, \rho^P_\Xi) &\geq \frac{d_{C(\Xi)}(b_x, \rho^P_\Xi) + d_{C(\Xi)}(b_y, \rho^P_\Xi) - d_{C(\Xi)}(b_x, b_y)}{2} \\ &\geq \frac{(C - 1 - D) + (C - 1 - D - 2\epsilon) - K}{2} = 1 \end{aligned}$$

Suppose there is a cone vertex v_Ψ on η, for some proper parabolic subcomplex Ψ of Ξ. If $P \leq \operatorname{Stab}(\Psi)$ then Ψ contains a P–fiber Υ'. That is impossible, because then $\Upsilon' \subset \rho^P_\Xi$, which would give a contradiction:

$$1 \leq d_{C(\Xi)}(v_\Psi, \rho^P_\Xi) \leq d_{C(\Xi)}(v_\Psi, \Upsilon') = 1/2$$

Thus, $P \not\leq \operatorname{Stab}(\Psi)$. By Lemma 2.2(5) and (6), $\operatorname{Stab}(\mathfrak{p}_\Upsilon(\Psi)) = \operatorname{Stab}(\Upsilon) \cap \operatorname{Stab}(\Psi) \leq \operatorname{Stab}(\Psi)$, so $\operatorname{Stab}(\mathfrak{p}_\Upsilon(\Psi)) \neq P$, so $\mathfrak{p}_\Upsilon(\Psi)$ spans a proper parabolic subcomplex of Υ. Thus, $P \not\leq \operatorname{Stab}(\Psi)$ implies $\operatorname{diam}_{\delta_\Upsilon}(\Psi) \leq 1$, since $\mathfrak{p}_\Upsilon(\Psi)$ is coned off in $C(\Upsilon)$.

Consider the sequence $b_x = z_0, z_1, \ldots, z_n = b_y$ of $\Sigma^{(0)}$ vertices along η. Each consecutive pair z_i, z_{i+1} either are mutually adjacent to a cone vertex v_Ψ of η, or are an adjacent pair in $\Sigma^{(1)}$. In the former case, the previous paragraph says $\delta_\Upsilon(z_i, z_{i+1}) \leq 1$. In the latter case, the fact that $\mathfrak{p}_\Upsilon$ is 1–Lipschitz, gives the same conclusion. Thus, $\mathfrak{p}_\Upsilon(\eta)$ has length in $C(\Upsilon)$ bounded above by the length of η in $C(\Xi)$, which was at most K. Now, one endpoint b_x of η sits on β_x, which is a combinatorial geodesic from x' to $x'' = \mathfrak{p}_\Upsilon(x')$, so $\mathfrak{p}_\Upsilon(b_x) = x''$. Similarly, the other endpoint b_y of η projects to y''. Thus, $\delta_\Upsilon(x'', y'') \leq K$. □

4.2.8. *Large links.* Let B and K be common BGI constants satisfying both [17, Corollary 5.25] and Proposition 4.5. Let $C := 2B + 3$. Assume $E > \max\{2K + 2C + 2, |S|/2\}$. If $Q \in \mathfrak{S}$ is an irreducible higher rank affine parabolic subgroup then it is $\sqsubseteq$–minimal, so there is nothing to prove.

Suppose $Q \in \mathfrak{S}$ is reducible, and write $Q = Q_1 \times \cdots \times Q_n$, where $2 \leq n \leq \lfloor |S|/2 \rfloor \leq E$ and the Q_i are nonspherical irreducible parabolic subgroups. Since Q is wide, each Q_i is pre-wide, by Lemma 2.1, so each Q_i is a domain that is properly nested in Q. For each i, $\delta_Q(x, \rho^{Q_i}_Q) = 0$, since $\mathcal{C}Q$ is a point. If $P \sqsubsetneq Q$ is reducible then $\mathcal{C}P$ is a point and $\delta_P(x, y) = 0$, so if $\delta_P(x, y) \geq E$ then P is irreducible. An irreducible parabolic subgroup of Q is contained in one of the factors Q_i. A similar proof applies for reducible $\bowtie$ by replacing Q by W^{ess}. This completes the verification of the Large Links Axiom for reducible domains.

Now suppose $Q \in \mathfrak{S}$ is irreducible and not higher rank affine. If $Q = \bowtie$ then $\mathcal{C}Q = \widehat{\Sigma}$. Otherwise, for any choice of $\Xi \in \operatorname{Stab}^{-1}(Q)$ there is a canonical identification of $\mathcal{C}Q$ with $C(\Xi)$. Let γ be a geodesic in $\mathcal{C}Q$ from $\pi_Q(x)$ to $\pi_Q(y)$. For each cone vertex v_Ψ of γ, we claim that every nonspherical irreducible factor in the canonical product decomposition of $\operatorname{Stab}(\Psi)$ is a domain. If there is more than one nonspherical irreducible factor, then those factors are pre-wide, by Lemma 2.1, hence are domains. If $\operatorname{Stab}(\Psi)$ has exactly one nonspherical irreducible factor R and $Q = \bowtie$ then for v_Ψ to have occurred as a cone vertex, R must be higher rank affine, so R is a domain. If $Q \neq \bowtie$ then $N_W(Q)$ is wide, so $N_W(R)$ is wide, hence R is pre-wide, by Lemma 2.1, so R is a domain. Take $\mathcal{R}$ to be all nonspherical irreducible factors of $\operatorname{Stab}(\Psi)$, as v_Ψ ranges over the cone vertices on γ. Then $|\mathcal{R}| \leq \delta_Q(x, y) \cdot \lfloor |S|/2 \rfloor$. This is less than N, since $E \geq \lfloor |S|/2 \rfloor$.

Each $R \in \mathcal{R}$ is properly nested in Q. This is clear if $Q = \bowtie$. If $Q \neq \bowtie$ then $R \leq \operatorname{Stab}(\Psi)$ for some cone vertex $v_\Psi \in \mathcal{C}Q = C(\Xi)$, so Ψ is proper parabolic subcomplex of Ξ, so $R \leq \operatorname{Stab}(\Psi) \lneq Q$.

We check the location of relative projections: if $R \in \mathcal{R}$ arose as a nonspherical irreducible factor of $\operatorname{Stab}(\Psi)$, then choose $\Upsilon \in \operatorname{Stab}^{-1}(R)$ with $\Upsilon \subset \Psi$. Then $\Upsilon \subset \rho^R_Q$ and every vertex of Υ is at distance $1/2$ from $v_\Psi \in \mathcal{C}Q$. Thus, $\delta_Q(x, \rho^R_Q) \leq \delta_Q(x, v_\Psi) + 1/2 \leq \delta_Q(x, y) + 1/2 \leq N$.

Finally, suppose $P \in \mathfrak{S}_Q \smallsetminus \{Q\}$ and $\delta_P(x, y) \geq E$. This implies P is irreducible, since otherwise $\delta_P(x, y) = 0$. Since $P \in \mathfrak{S}$ is irreducible it is pre-wide. Since $E > K$, BGI implies γ passes within distance B of ρ^P_Q in $\mathcal{C}Q$. Let $p \in \rho^P_Q$ and $q \in \gamma$ be points such that $\delta_Q(p, q) \leq B$. If $\delta_Q(x, q) < C$, set $u := \pi_Q(x)$. Otherwise, take u to be a vertex of $\Sigma^{(0)}$ on the subsegment of γ between $\pi_Q(x)$ and q such that $C \leq \delta_Q(q, u) \leq C + 1$. Similarly, take v to be a $\Sigma^{(0)}$ vertex on the subsegment of γ between q and $\pi_Q(y)$ that is either at distance between C

and $C+1$ from q, or equal to $\pi_Q(y)$ if the subsegment is too short. If γ_{uv} is the segment of γ from u to v:

$$|\gamma_{uv}|_{\mathcal{C}Q} \leq 2C+2 \tag{2}$$

Suppose $u \neq \pi_Q(x)$. For every point u' on the subsegment of γ between $\pi_Q(x)$ and u and every $p' \in \rho_Q^P$:

$$\delta_Q(u',p') \geq \delta_Q(u',q) - \delta_Q(q,p) - \delta_Q(p,p') \geq C - B - 1 > B$$

Thus, BGI gives $\delta_P(x,u) \leq K$ when $u \neq \pi_Q(x)$. If $u = \pi_Q(x)$ then $\delta_P(x,u) = 0 \leq K$.

Symmetrically, $\delta_P(y,v) \leq K$. Thus:

$$\delta_P(u,v) \geq \delta_P(x,y) - 2K \geq E - 2K > 2C+2 \tag{3}$$

Now we claim that $P \leq \mathrm{Stab}(\Psi)$ for some cone vertex v_Ψ of γ_{uv}. Otherwise, as in the proof of Proposition 4.5, we find that γ_{uv} has δ_P–diameter at most equal to its $\mathcal{C}Q$–length, which gives a contradiction between (2) and (3). Thus, $P \leq \mathrm{Stab}(\Psi)$. Since P is nonspherical and irreducible, it is contained in some nonspherical irreducible factor R of the canonical product decomposition of $\mathrm{Stab}(\Psi)$. By construction, $R \in \mathcal{R}$, so this completes the verification of the Large Links Axiom.

4.2.9. *Partial realization.* In Coxeter groups the Partial Realization Axiom, with $E = 0$, follows easily from:

Lemma 4.6. *If $P_1,\dots,P_n$ are pairwise commuting essential parabolic subgroups then there exist $w \in W$ and disjoint $T_i \subset S$ such that $P_i = wW_{T_i}w^{-1}$ for all i.*

Proof. Since P_i is essential, its canonical product decomposition contains only nonspherical irreducible components. These commute with each other and with every other P_j, so by replacing each P_i with its factors, we may assume that the original collection contains only nonspherical irreducible parabolic subgroups.

The proof is by induction on n. Since P_1 is parabolic, there exist $v \in W$ and $T_1 \subset S$ such that $P_1 = vW_{T_1}v^{-1}$. If $n = 1$ we are done, by taking $w := v$. Otherwise, since P_1 is nonspherical and irreducible, $Z_W(P_1)$ is the parabolic subgroup $P_1^\perp = vW_{T_1^\perp}v^{-1}$. Since the subgroups $P_2,\dots,P_n$ commute with P_1, they are all in $P_1^\perp$, so for $i > 1$ we have that $v^{-1}P_2v,\dots,v^{-1}P_nv$ is a collection of $n-1$ pairwise commuting parabolic subgroups of $W_{T_1^\perp}$. By induction, there is a common conjugator $u \in W_{T_1^\perp}$ and disjoint subsets $T_2,\dots,T_n \subset T_1^\perp \subset S$ such that $uv^{-1}P_ivu^{-1} = W_{T_i}$ for all $2 \leq i \leq n$. Since $u \in W_{T_1^\perp}$, we have $uv^{-1}P_1vu^{-1} = uW_{T_1}u^{-1} = W_{T_1}$. This proves the lemma, by taking $w := vu^{-1}$. □

Suppose $P_1,\dots,P_n$ is a collection of pairwise orthogonal domains. If some $P_i = \mathsf{W}$ then $i = n = 1$, since no domain is orthogonal to W. If W is reducible then $\mathcal{C}\mathsf{W}$ is a point, so any x works. If W is irreducible then $\mathcal{C}\mathsf{W} = \widehat{\Sigma}$. Every $p \in \pi_{\mathsf{W}}(\Sigma^{(0)})$ is itself a vertex in $\Sigma^{(0)}$, since the projection map in this case is just the identity on $\Sigma^{(0)}$, so take $x = p$. Then $\delta_{\mathsf{W}}(p,x) = 0$. There are no domains transverse to W or in which W is properly nested, so the second realization condition is vacuous for W.

Otherwise, all P_i are essential parabolic subgroups. By Lemma 4.6, there exist $w \in W$ and disjoint $T_i \subset S$ such that $P_i = wW_{T_i}w^{-1}$ for all i. For each i set $\Upsilon_i := w\Sigma_{T_i} \in \mathrm{Stab}^{-1}(P_i)$. If P_i is irreducible then $\mathcal{C}P_i$ is canonically isomorphic to $C(\Upsilon_i)$ and $\pi_{P_i}(\Sigma^{(0)})$ is identified with $\Upsilon_i^{(0)}$, so p_i is a vertex of $\Upsilon_i \subset \Sigma$, hence $p_i = wa_i$ for some $a_i \in W_{T_i}$. If P_i is reducible set $a_i := 1$. Define $x = wa_1\cdots a_n \in W$. The a_i all commute, since the subgroups W_{T_i} pairwise commute, so $b_i := a_1\cdots a_na_i^{-1} \in W_{T_i^\perp}$. Let $\Xi_i := wb_i\Sigma_{T_i} \in \mathrm{Stab}^{-1}(P_i)$. Then we have $x = wb_ia_i \in \Xi_i$, and we use this to verify the two realization conditions.

If P_i is reducible then Condition 9(a) is immediate from the fact that $\mathcal{C}P_i$ is a single point, so suppose P_i is irreducible. We have points in two P_i–fibers, $x \in \Xi_i$ and $p_i = wa_i \in \Upsilon_i$. But $x = wb_iw^{-1}p_i$, so Lemma 2.2(4) says $\mathfrak{p}_{\Upsilon_i}(x) = \mathfrak{p}_{\Upsilon_i}(p_i) = p_i$. Thus, $\delta_{P_i}(p_i,x) = 0$.

For Condition 9(b), if $Q \in \mathfrak{S}$ with $(P_i \sqsubsetneq Q) \vee (P_i \pitchfork Q)$ then $\pi_Q(x) \in \pi_Q(\Xi_i) \subset \rho_Q^{P_i}$, so $\delta_Q(x,\rho_Q^{P_i}) = 0$.

4.2.10. *Uniqueness.* Fix r. Take $M(r) := M(r,\dots,r)$ as in [17, Proposition 5.23].

Suppose that γ is a combinatorial geodesic in $\Sigma^{(1)}$ between $x,y \in \Sigma^{(0)}$, with $d_1(x,y) > M(r)$.

If γ contains a subsegment γ' of length greater than $M(r)$ that is contained in a wide parabolic subcomplex Υ then every wall dual to an edge of γ' cuts Υ and separates x and y, by Lemma 2.2(2), so $d_\Upsilon(x,y) \geq |\gamma'| > M(r)$. By [17, Proposition 5.23], there exists a pre-wide $P \leq \mathrm{Stab}(\Upsilon)$ such that $\delta_P(x,y) > r$. We have already excluded the case that W^{ess} is higher rank affine, so every pre-wide parabolic subgroup belongs to $\mathfrak{S}$. This case necessarily holds when W is wide, since then γ is contained in the wide parabolic subcomplex Σ.

If no subsegment of γ of length at least $M(r)+1$ is contained in a wide parabolic subcomplex, then [17, Theorem 3.31] says γ is Morse, with Morse gauge depending only on $M(r)$, and then [17, Corollary 4.20] says that the inclusion of γ into $\widehat{\Sigma} = \mathcal{C}\mathsf{W}$ is quasigeodesic, with quasigeodesic constants determined by $M(r)$. In particular, $\delta_{\mathsf{W}}(x,y)$ is bounded below by a linear function of $d_1(x,y)$, with constants determined by $M(r)$. Thus, $\delta_{\mathsf{W}}(x,y) \geq r$ once $d_1(x,y)$ is sufficiently large.

4.3. **Verifying coarse equivariance.** So far we have verified that $\Sigma^{(1)}$, which is the Cayley graph of W with respect to S, is a relatively hierarchically hyperbolic *space*. To say that W is a relatively hierarchically hyperbolic *group*, we must verify the additional requirements of Definition 2.19. Define the action of W on $\mathfrak{S}$ as follows. For all $w \in W$ and $P \in \mathfrak{S} \smallsetminus \{\mathsf{W}\}$, define $w.\mathsf{W} := \mathsf{W}$ and $w.P := wPw^{-1}$. Cofiniteness follows because each $P \in \mathfrak{S} \smallsetminus \{\mathsf{W}\}$ is conjugate to a special subgroup W_T, and there are only finitely many subsets $T \subset S$. Conjugation of parabolic subgroups preserves inclusion and commutation, so the action on $\mathfrak{S}$ preserves nesting and orthogonality.

The coordinate isometries are induced by the isometric action of W on Σ. For reducible $P \in \mathfrak{S}$ there is nothing to prove, since the coordinate space $\mathcal{C}P$ is a single point. For irreducible P and $\Upsilon \in \mathrm{Stab}^{-1}(P)$, let $\phi_\Upsilon \colon C(\Upsilon) \to \mathcal{C}P$ be the canonical isomorphism. Since the g–action on Σ takes parabolic subcomplexes to parabolic subcomplexes, it induces an isomorphism $g^*_\Upsilon \colon C(\Upsilon) \to C(g\Upsilon)$. Define $g_P \colon \mathcal{C}P \to \mathcal{C}g.P$: $\phi_{g\Upsilon} \circ g^*_\Upsilon \circ \phi^{-1}_\Upsilon$. This is independent of the choice of Υ, because any other element of $\mathrm{Stab}^{-1}(P)$ is of the form $z\Upsilon$ for $z \in P^\perp$. But $\phi_{z\Upsilon} = \phi_\Upsilon \circ (z^{-1})^*_{z\Upsilon}$ and $\phi_{gz\Upsilon} = \phi_{g\Upsilon} \circ (gz^{-1}g^{-1})^*_{gz\Upsilon}$. Thus:

$$\begin{aligned}\phi_{gz\Upsilon} \circ g^*_{z\Upsilon} \circ \phi^{-1}_{z\Upsilon} &= \phi_{g\Upsilon} \circ (gz^{-1}g^{-1})^*_{gz\Upsilon} \circ g^*_{z\Upsilon} \circ z^*_\Upsilon \circ \phi^{-1}_\Upsilon \\ &= \phi_{g\Upsilon} \circ g^*_\Upsilon \circ \phi^{-1}_\Upsilon\end{aligned}$$

Composability of these isometries follows because for $\Upsilon \in \mathrm{Stab}^{-1}(P)$ we have $h\Upsilon \in \mathrm{Stab}^{-1}(hPh^{-1}) = \mathrm{Stab}^{-1}(h.P)$ and $gh\Upsilon \in \mathrm{Stab}^{-1}(ghPh^{-1}g^{-1}) = \mathrm{Stab}^{-1}((gh).P)$, so:

$$\begin{aligned}(gh)_P &:= \phi_{gh\Upsilon} \circ (gh)^*_\Upsilon \circ \phi^{-1}_\Upsilon \\ &= \phi_{gh\Upsilon} \circ g^*_{h\Upsilon} \circ \phi^{-1}_{h\Upsilon} \circ \phi_{h\Upsilon} \circ h^*_\Upsilon \circ \phi^{-1}_\Upsilon \\ &= g_{h.P} \circ h_P\end{aligned}$$

Similar arguments apply to irreducible W, since the W–action on Σ induces an isometric action on $\widehat{\Sigma} = \mathcal{C}\mathsf{W}$.

The coarse equivariance properties of Definition 2.19 likewise follow from tracing through the definitions and using the W–action on Σ. In fact, they are equivariant on the nose: we can take $E = 0$.

5. Promoting the Coxeter relative HHG structure

In this section we extend the relative HHG structure provided by Theorem 1.1. As described in the introduction, we have two different approaches to arranging the necessary equivariance conditions, one by passing to a finite index subgroup of W and one by restricting the affine parabolics.

We start by choosing a finite index subgroup of W and showing it admits an HHG structure. The structure has an additional property that is needed to deduce Theorem 1.7 from Theorem 1.4:

Definition 5.1 ([37])**.** An HHG $(G, \mathfrak{S})$ is *colorable* if there exists an assignment of finitely many colors to the domains of $\mathfrak{S}$ such that every pair of distinct domains with the same color is transverse, and such that the G–action on $\mathfrak{S}$ induces a permutation action of G on the set of colors.

In the proof of Theorem 1.4 we start with a relative HHS structure $\mathfrak{S}$ whose peripheral coordinate spaces are HHSs, build a "proto-HHS structure" $\mathfrak{S}^+$ by replacing the peripheral domains by their corresponding coordinate HHS structures, and then complete the proto-HHS to a full HHS structure $\mathfrak{S}^{++}$ by adding "dummy container domains". This is a general relative HHS promotion recipe, whose details appear in Appendix A. The main work is to arrange that the peripheral stabilizers respect their peripheral HHS structures, so that the promotion can be carried out equivariantly, as in Proposition A.5. We achieve this by passing to a congruence subgroup of W. Separate arguments establish cofiniteness of the action on $\mathfrak{S}^{++}$ and colorability.

Let (W,S) be a finite rank Coxeter system with $S = \{s_1, \dots, s_n\}$ and Coxeter matrix $M := (m_{ij})$. Let $C := (-\cos(\pi/m_{ij})) = (c_{ij})$ be its cosine matrix. The *Tits form* B_M on $V := \mathbb{R}^{|S|}$ is the W–invariant

symmetric bilinear form $B_M(\mathbf{e}_i, \mathbf{e}_j) = c_{ij}$, where $\mathbf{e}_i$ is the basis vector of V corresponding to $s_i \in S$. The *Tits representation*[1] is the faithful representation $\rho\colon W \to \mathrm{GL}(V)$ defined by $\rho(s_i) := \mathbf{v} \mapsto \mathbf{v} - 2B_M(\mathbf{v}, \mathbf{e}_i)\mathbf{e}_i$.

Let $R := \mathbb{Z}[2c_{ij} \mid 1 \leq i, j \leq n]$ be the ring generated by the finitely many numbers $2c_{ij}$. The numbers $2c_{ij}$ are algebraic integers, so R is a finitely generated $\mathbb{Z}$–module, and an order in the number field $\mathbb{Q}(2c_{ij} \mid 1 \leq i, j \leq n)$ contained in its ring of integers. Thus, for any nontrivial ideal $\mathcal{I}$ in R, for example a principal ideal $\mathcal{I} = (a) = \{ar \mid r \in R\}$ generated by some $a \in R$, the quotient $R/\mathcal{I}$ is a finite ring. The matrix of $\rho(s_i)$ has entries in R, so we get a finite index normal subgroup of W:

$$K_{\mathcal{I}} := \ker(W \xrightarrow{\rho} \mathrm{GL}_{|S|}(R) \to \mathrm{GL}_{|S|}(R/\mathcal{I})) \tag{4}$$

The following argument is extracted from the proof of the well-known result that arithmetic groups have a finite index "neat" subgroup. We give a proof for completeness.

Lemma 5.2. *Let p be an odd prime. Elements of $\ker(\mathrm{GL}_n(R) \to \mathrm{GL}_n(R/(p)))$ have no nontrivial root of unity eigenvalues.*

Proof. Suppose $A \in \ker(\mathrm{GL}_n(R) \to \mathrm{GL}_n(R/(p)))$. There exists $B \in M_n(R)$ such that $A = I_n + pB$. If ζ is an eigenvalue of A then $\beta := \frac{1-\zeta}{p} \in \mathbb{Q}(\zeta)$ is an eigenvalue of $-B$. Since β is integral over R, which is integral over $\mathbb{Z}$, β is an algebraic integer, so its field norm $N_{\mathbb{Q}(\zeta)/\mathbb{Q}}(\beta)$ is an integer. Multiplicativity of the norm gives:

$$1 - \zeta = p\beta \implies N_{\mathbb{Q}(\zeta)/\mathbb{Q}}(1-\zeta) = N_{\mathbb{Q}(\zeta)/\mathbb{Q}}(p) N_{\mathbb{Q}(\zeta)/\mathbb{Q}}(\beta)$$

Thus, $N_{\mathbb{Q}(\zeta)/\mathbb{Q}}(p)$ divides $N_{\mathbb{Q}(\zeta)/\mathbb{Q}}(1-\zeta)$. If ζ is a primitive m–th root of unity, for some $m > 1$, then $N_{\mathbb{Q}(\zeta)/\mathbb{Q}}(p) = p^{\varphi(m)}$ divides $N_{\mathbb{Q}(\zeta)/\mathbb{Q}}(1-\zeta) = \Phi_m(1)$. Since $\Phi_m(1)$ is 1 or prime, we must have $\varphi(m) = 1$, which implies $m = 2$. But $\Phi_2(1) = 2$, so $p = 2$, contradicting that p is odd. □

Let $P = W_T$ be an irreducible affine special subgroup of (W, S). It splits as $W_T = \Lambda_{W_T} \rtimes F_{W_T}$, where $\Lambda_{W_T} \cong \mathbb{Z}^{|T|-1}$ is the translation subgroup and F_{W_T} is the finite linear part. In crystallographic group theory F_{W_T} is called the *point group*. If $V_T \leq V$ is the subspace spanned by the basis vectors corresponding to elements of T, the restriction of B_M to V_T is positive semidefinite with one-dimensional radical, [18, Theorem 6.8.12(ii)]. The W_T–action preserves the radical. The $W_{T^\perp}$–action fixes V_T pointwise. Thus, the restriction of the Tits representation to $N_W(W_T) = W_T \times W_{T^\perp}$ descends to a representation:

$$\bar{\rho}\colon N_W(W_T) \to \mathrm{GL}(V_T/\operatorname{rad}(V_T))$$

Moreover, $\ker(\bar{\rho}) = \Lambda_{W_T} \times W_{T^\perp}$ and $\bar{\rho}(N_W(W_T)) \cong F_P$.

Lemma 5.3. *Let (W, S) be a finite rank Coxeter system, and let $\mathfrak{P}$ be its set of irreducible higher rank affine parabolic subgroups. For any odd prime p, let $K_{(p)} \leq W$ as in (4). For every $P \in \mathfrak{P}$:*

$$K_{(p)} \cap N_W(P) \leq \ker(N_W(P) \to N_W(P)/P^\perp \cong P \to P/\Lambda_P = F_P)$$

In particular, $K_{(p)} \cap N_W(P) \leq \Lambda_P \times P^\perp$.

Proof. Since $K_{(p)}$ is normal, we may assume $P = W_T$ is special. Suppose $g = \tau f z \in K_{(p)} \cap N_W(W_T)$ with $\tau \in \Lambda_{W_T}$, $f \in F_{W_T}$, and $z \in W_{T^\perp}$. Then $\bar{\rho}(g) = \bar{\rho}(f)$ is a finite order matrix. If f has order m then every eigenvalue λ of $\bar{\rho}(g)$ satisfies $\lambda^m = 1$. Suppose $m > 1$. Then $\bar{\rho}(g) = \bar{\rho}(f)$ is a nontrivial matrix of finite order m dividing $|F_{W_T}|$. Since $\bar{\rho}(g)$ is nontrivial, not all of its eigenvalues are 1, so it has some eigenvalue of order $d > 1$ with d dividing m. The flag $0 < \operatorname{rad}(V_T) < V_T < V$ is $N_W(W_T)$–invariant, so every eigenvalue of $\bar{\rho}(g)$ is an eigenvalue of $\rho(g)$. On the other hand, for $g \in K_{(p)}$, Lemma 5.2 says $\rho(g)$ has no nontrivial root of unity eigenvalues. Thus, $f = 1$, so $g \in \Lambda_{W_T} \times W_{T^\perp}$. □

Proof of Theorem 1.4. If W is hyperbolic it is already an HHG. If W^{ess} is irreducible higher rank affine then W is virtually Abelian, so it is virtually an HHG. Both of these cases are trivially colorable. Assume we are not in one of these cases. Then the HHG structure $\mathfrak{S}$ for W relative to the set $\mathfrak{P}$ of the irreducible higher rank affine parabolic subgroups of (W, S) given by Theorem 1.1 has $\mathfrak{S} \neq \{\mathfrak{W}\}$.

For $P \in \mathfrak{P}$, let $\Lambda_P \cong \mathbb{Z}^{\mathrm{rank}(P)-1}$ be its translation subgroup and let F_P be its point group. Let $\mathfrak{Q}$ be the special subgroups in $\mathfrak{P}$. Pick any odd prime p and take the congruence subgroup $K_{(p)} \leq W$ as defined above.

Since $K_{(p)}$ is a finite index subgroup of W, the relative HHG structure on W transfers to a relative HHG structure on $K_{(p)}$ via the $K_{(p)}$–equivariant quasiisometry $K_{(p)} \hookrightarrow W$. Our goal is to choose HHS structures

[1]The Tits representation is called the "canonical representation" in [18] and the "geometric representation" in [10].

on representatives of $K_{(p)}$–orbits of peripherals, and show that the $K_{(p)}$–stabilizers of those representatives satisfy the hypotheses of Proposition A.5.

Since the action of W on $\mathfrak{P}$ is by conjugation, $\mathrm{Stab}_W(P) = N_W(P)$. Thus, Lemma 5.3 gives $\mathrm{Stab}_{K_{(p)}}(P) = K_{(p)} \cap N_W(P) \leq \Lambda_P \times P^\perp$. If $g \in \mathrm{Stab}_{K_{(p)}}(P)$ then we can write $g = yz$ for $y \in \Lambda_P$ and $z \in P^\perp$. Since $P^\perp$ acts trivially on $\mathcal{C}P = C(P)$, the action of g on $\mathcal{C}P$ is identical to the action of y on $\mathcal{C}P$. Thus, the $\mathrm{Stab}_{K_{(p)}}(P)$–action on $\mathcal{C}P$ factors through the Λ_P–action.

For each $K_{(p)}$–orbit in $\mathfrak{P}$, choose a representative P. Define the standard product HHS structure on $\Lambda_P \cong \mathbb{Z}^{\mathrm{rank}(P)-1}$, obtained as the product of $\mathrm{rank}(P) - 1$ $\mathbb{Z}$ factors. This structure is finite and Λ_P–invariant. There are only finitely many such P, because there are finitely many W–orbits and $[W : K_{(p)}] < \infty$.

Since Λ_P has finite index in P and P is equivariantly quasiisometric to $C(P) = \mathcal{C}P$, an orbit map q of Λ_P into $\mathcal{C}P$ is a Λ_P–equivariant quasiisometry. Let $\mathfrak{T}_P$ be the HHS structure on $\mathcal{C}P$ obtained by transferring the product HHS structure on Λ_P via the quasiisometry; that is, keep the same domains and coordinate spaces, and pre-compose projections with a chosen inverse quasiisometry $\bar{q}$. Since q is Λ_P–equivariant, $\bar{q}$ is coarsely Λ_P–equivariant. Moreover, we have only used finitely many P, so the HHS constants are uniform.

Now we can verify the hypotheses of Proposition A.5. Hypothesis (1) is true because the $\mathrm{Stab}_{K_{(p)}}(P)$–action on $\mathfrak{T}_P$ factors through the Λ_P–action, which fixes every domain of $\mathfrak{T}_P$. If $g = yz \in \mathrm{Stab}_{K_{(p)}}(P)$ for $y \in \Lambda_P$ and $z \in P^\perp$ then we define the isometry $g_U \colon \mathcal{C}U \to \mathcal{C}g.U = \mathcal{C}U$ required by (2) as follows. If $\mathcal{C}U$ is a single point then g_U is the identity map. Otherwise, $\mathcal{C}U = \mathbb{R}$ and g_U is the translation given by the U–coordinate of y. Hypothesis (3) follows immediately.

To see that projections are coarsely $\mathrm{Stab}_{K_{(p)}}(P)$–equivariant, hypothesis (4), recall that for domain U with nonsingleton coordinate space, the projection π_U was defined by precomposing a coordinate projection $\pi_U^{\Lambda_P}$ in Λ_P with the quasiisometry $\bar{q}\colon \mathcal{C}P \to \Lambda_P$. Thus, $g_U \circ \pi_U = y_U \circ \pi_U^{\Lambda_P} \circ \bar{q} = \pi_U^{\Lambda_P} \circ y \circ \bar{q}$, while $\pi_{g.U} \circ g_P = \pi_U^{\Lambda_P} \circ \bar{q} \circ y_P$. Since $\bar{q}$ is coarsely Λ_P–equivariant, these two uniformly coarsely agree.

Hypothesis (5) is trivial, because in $\mathfrak{T}_P$ only the $\sqsubseteq$–minimal domains have nonsingleton coordinate spaces, and those domains are pairwise orthogonal, so $(U \sqsubsetneq V) \vee (U \pitchfork V)$ implies $\mathcal{C}V$ is a point.

Proposition A.5 gives a $K_{(p)}$–HHS structure $\mathfrak{S}^{++}$ on $K_{(p)}$ built from the restriction of $\mathfrak{S}$ to $K_{(p)}$, promoted to a proto-HHS structure $\mathfrak{S}^+$ by adding the standard product HHG structures on the translation subgroups of the irreducible higher rank affine parabolic subgroups, and then completed to $\mathfrak{S}^{++}$ by adding dummy container domains. To conclude that $\mathfrak{S}^{++}$ is actually an HHG structure, not just a $K_{(p)}$–HHS, we show that the $K_{(p)}$–action satisfies the hypothesis of Corollary A.4.

The $K_{(p)}$–action on $\mathfrak{S}^+$ is cofinite, so we show that for each $R \in \mathfrak{S}^+$ the $\mathrm{Stab}_{K_{(p)}}(R)$–action on the set of nonmaximal sets $\mathfrak{O}$ of pairwise orthogonal domains in $\mathfrak{S}^+_R \smallsetminus \{R\}$ is cofinite.

The peripheral HHG structures $\mathfrak{T}_P$ are finite, so if $R \in \mathfrak{T}_P$ we are done, so we may assume $R \in \mathfrak{S} \smallsetminus \mathfrak{P}$. Let $\{Q_1, \dots, Q_n\} := \mathfrak{O} \cap (\mathfrak{S} \smallsetminus \mathfrak{P})$. Note that $Q_j \sqsubsetneq R$, so $Q_j \neq \mathsf{W}$, so Q_j is some essential parabolic subgroup of W. Let $P_1, \dots, P_m$ be the peripheral domains such that $\mathfrak{O}$ contains some domain from $\mathfrak{T}_{P_i}$. Then $\mathfrak{O}' := \{P_1, \dots, P_m, Q_1, \dots, Q_n\}$ is a family of pairwise commuting essential parabolic subgroups of W. Since the $\mathfrak{T}_P$ are finite, the map $\mathfrak{O} \to \mathfrak{O}'$ is finite-to-one, so it suffices to show that $\mathrm{Stab}_{K_{(p)}}(R)$ acts cofinitely on sets of the form $\mathfrak{O}'$.

If $R = \mathsf{W}$ then $\mathrm{Stab}_{K_{(p)}}(R) = K_{(p)}$, and Lemma 4.6, says $\mathfrak{O}'$ can be simultaneously conjugated in W to a collection of pairwise commuting special subgroups. There are only finitely many special subgroups, so only finitely many subsets of pairwise commuting ones, so there are finitely many W–orbits of sets of the form of $\mathfrak{O}'$. Since $K_{(p)}$ has finite index in W, there are also finitely many $K_{(p)}$–orbits.

If $R \neq \mathsf{W}$ then $R \leq \mathrm{Stab}_W(R)$. The argument is the same as in the W–case, applying Lemma 4.6 with R as the ambient Coxeter group. Since $K_{(p)} \cap R$ has finite index in R, the $\mathrm{Stab}_{K_{(p)}}(R)$–action is also cofinite.

Finally, we show that $\mathfrak{S}^{++}$ is colorable. We will actually produce a $K_{(p)}$–invariant coloring, which is stronger than what is required by Definition 5.1. Fix the power set of S as one color palette. If $W^{ess} = W_T$ give W color T. For $P \in \mathfrak{S} \smallsetminus (\{\mathsf{W}\} \cup \mathfrak{P})$, color P the unique subset $T \subset S$ such that P is conjugate to W_T. If P and P' are both conjugate to W_T then they are not properly nested, since then they would have different ranks, and they do not commute, because if they did then P' would be conjugate into $W_{T^\perp}$, but it cannot be since the unique special subgroup to which it is conjugate is W_T and $T \cap T^\perp = \emptyset$. Thus, if P and $P' \in \mathfrak{S} \smallsetminus (\{\mathsf{W}\} \cup \mathfrak{P})$ have the same color then they are transverse.

Now consider the domains of $\mathfrak{S}^+ \smallsetminus \mathfrak{S}$. They come from $\mathfrak{T}_P$ for some $P \in \mathfrak{P}$. The $\mathfrak{T}_P$ are finite. Take disjoint color palettes for the distinct $K_{(p)}$–orbits in $\mathfrak{P}$, such that for a chosen representative P of each $K_{(p)}$

orbit of $\mathfrak{P}$, we can assign every domain in $\mathfrak{T}_P$ its own color. Since the $\mathrm{Stab}_{K_{(p)}}(P)$–action on $\mathfrak{T}_P$ factors through the Λ_P–action, which is trivial on $\mathfrak{T}_P$, the $\mathrm{Stab}_{K_{(p)}}(P)$–action is color preserving. Thus, there is a well-defined $K_{(p)}$–invariant coloring on the entire union $\bigcup_{w\in K_{(p)}} w\mathfrak{T}_P$. Moreover, distinct domains have the same color if and only if they are $w.U$ and $w'.U$ for some $U \in \mathfrak{T}_P$ and $w, w' \in K_{(p)}$ with $w.P \neq w'.P$. By the argument of the previous paragraph, $w.P \pitchfork w'.P$, so by definition of $\mathfrak{S}^+$, we also have $w.U \pitchfork w'.U$.

The remaining domains are the dummy container domains, indexed by pairs $(R, \mathfrak{O})$ for $R \in \mathfrak{S}^+$ and $\mathfrak{O}$ a nonempty, nonmaximal set of pairwise orthogonal domains of $\mathfrak{S}^+_R \smallsetminus \{R\}$. Color these by a pair whose first entry is the color of R and whose second entry is the set of colors occurring in $\mathfrak{O}$. These are different from all of the colors assigned to $\mathfrak{S}^+$, so it remains only to check that if $(R, \mathfrak{O})$ and $(R', \mathfrak{O}')$ have the same color then they are transverse. Note that since elements of $\mathfrak{O}$ are pairwise orthogonal they have distinct colors.

Consider the orthogonality and nesting relations for dummy container domains as constructed in [1, Theorem A.1]. $(R, \mathfrak{O})$ and $(R', \mathfrak{O}')$ are orthogonal if and only if $R \perp R'$, but we already know that if R and R' have the same color they are either equal or transverse, so if $(R, \mathfrak{O})$ and $(R', \mathfrak{O}')$ have the same color they are not orthogonal. Suppose $(R, \mathfrak{O}) \sqsubseteq (R', \mathfrak{O}')$ and they have the same color. The nesting is true if and only if $R \sqsubseteq R'$ and for every $U' \in \mathfrak{O}'$ either $U' \perp R$ or $U' \sqsubseteq U$ for some $U \in \mathfrak{O}$. Since $(R, \mathfrak{O})$ and $(R', \mathfrak{O}')$ have the same color, so do R and R', which they cannot if $R \sqsubsetneq R'$, so $R = R'$. Then for every $U' \in \mathfrak{O}'$ there exists $U, V \in \mathfrak{O}$ with $U' \sqsubseteq U$ and V of the same color as U'. If $U \neq V$ then $U \perp V$, by construction. But then $U' \sqsubseteq U$ and $U \perp V$ implies $U' \perp V$, which contradicts that domains with the same color are transverse. Thus, $U = V$, which further implies $U' = U$, since U' cannot have the same color as U if it is properly nested. Since this was true for every $U' \in \mathfrak{O}'$, we have $\mathfrak{O} = \mathfrak{O}'$. Thus, distinct nested $(R, \mathfrak{O})$ and $(R', \mathfrak{O}')$ cannot have the same color. Conclude that two dummy container domains with the same colors are transverse. Moreover, the coloring is $K_{(p)}$–invariant since its component colorings were. □

The second idea we pursue is to improve the HHG structure on W relative to all irreducible higher rank affine parabolic subgroups to an HHG structure relative to the nonhyperoctahedral irreducible affine parabolic subgroups. This is Theorem 1.2. Theorem 1.3 shows that this is the best we can do; the presence of a nonhyperoctahedral irreducible affine parabolic subgroup obstructs the existence of an HHG structure on W. Theorem 1.2 follows by a similar argument to Theorem 1.4:

Proof of Theorem 1.2. Let $\mathfrak{P}_0$ and $\mathfrak{P}_1$ be the sets of irreducible higher rank affine parabolic subgroups of (W, S) that are nonhyperoctahedral and hyperoctahedral, respectively. These sets are invariant under W–conjugation, so $\mathfrak{P} = \mathfrak{P}_0 \sqcup \mathfrak{P}_1$ is a W–invariant decomposition of the peripheral domains of the relative HHG structure $\mathfrak{S}$ of Theorem 1.1.

If W is hyperbolic then it has no higher rank affine parabolic subgroups, so $\mathfrak{P} = \emptyset$ and $\mathfrak{S}$ is already an HHG structure for W, so we are done.

Suppose $P \in \mathfrak{P}_1$. Since P is affine, it acts as a Euclidean reflection group on a Euclidean space $\mathbb{E}^n$, with $n = \mathrm{rank}(P) - 1 > 1$. Since P is hyperoctahedral, there is a choice of coordinates identifying $\mathbb{R}^n$ with $\mathbb{E}^n$ such that the P–action is by translation followed by a signed permutation of the coordinates of $\mathbb{R}^n$. Let $\mathfrak{T}_P^{prod}$ be the standard product HHS structure on $E_P := (\mathbb{R}^n, \ell^1)$: its domains are indexed by nonempty subsets $I \subset \{1, \dots, n\}$ with nesting given by inclusion and orthogonality given by disjointness. The nesting minimal domains, indexed by singletons $\{i\}$, have coordinate space $\mathbb{R}$ with projection the i–th coordinate projection in $\mathbb{R}^n$. All other coordinate spaces are singletons. The HHS structure $\mathfrak{T}_P^{prod}$ makes P an HHG.

If W^{ess} is irreducible higher rank affine then we are done: if W^{ess} is hyperoctahedral then its product HHS gives an HHG structure on W, and if not then we have a trivial relative HHG structure for W whose unique domain is peripheral.

We proceed, assuming W is not hyperbolic and has essential part that is not irreducible affine, so $\mathfrak{S} \neq \{\mathsf{W}\}$.

Let $\mathfrak{Q}$ be the set of special subgroups in $\mathfrak{P}_1$, which accounts for exactly one representative of each W–orbit in $\mathfrak{P}_1$. For each $P \in \mathfrak{Q}$, let $\mathfrak{T}_P^{prod}$ be the product HHG structure on E_P described above. The orbit maps of P on E_P and $\mathcal{C}P = C(P)$ induce a coarsely P–equivariant quasiisometry $\phi_P \colon \mathcal{C}P \to E_P$. Let $\mathfrak{T}_P$ denote the HHS structure on $\mathcal{C}P$ pulled back from the structure $\mathfrak{T}_P^{prod}$ on E_P via ϕ_P. Since there are only finitely many conjugacy classes of parabolic subgroup in W, the set $\mathfrak{Q}$ is finite, so the $\mathfrak{T}_P$ have uniform HHS parameters.

$\mathrm{Stab}_W(P) = N_W(P) = P \times P^\perp$, and the $P^\perp$–action on $\mathcal{C}P$ is trivial. Thus, $\mathrm{Stab}_W(P)$ acts on $\mathfrak{T}_P$ by factoring through the P–action. Unlike in the proof of Theorem 1.4, the $\mathrm{Stab}_W(P)$–action is not purely translational, but we used hyperoctahedrality to choose E_P such that the linear component of the P–action on

$\mathfrak{T}_P^{prod}$ is by signed permutations of the coordinate axes, so we still get the desired permutations of the domains of $\mathfrak{T}_P^{prod}$ and corresponding isometries between its coordinate spaces. Since ϕ_P is coarsely P–equivariant, the desired properties also hold for $\mathfrak{T}_P$. The rest of the proof proceeds exactly as in the proof of Theorem 1.4, using Proposition A.5 to propagate the chosen $\mathfrak{T}_P$ for $P \in \mathfrak{Q}$ to all of $\mathfrak{P}_1$ in a compatible way, and then using partial promotion to get a new structure $\mathfrak{S}^{++}$ relative to $\mathfrak{P}_0$. Cofiniteness of the W–action on $\mathfrak{S}^{++}$ follows by the same argument as in Theorem 1.4. Thus, $\mathfrak{S}^{++}$ is an HHG structure on W relative to $\mathfrak{P}_0$. □

Azuelos and Hagen [4, Definition 7.1] define a virtually Abelian subgroup A of G to be *highest* if A is virtually $\mathbb{Z}^n$ and no finite index subgroup of A is contained in a subgroup of G isomorphic to $\mathbb{Z}^{n+1}$.

The proof of Theorem 1.3 hinges on the following Coxeter nonhyperoctahedral escalation lemma:

Lemma 5.4. *Every nonhyperoctahedral irreducible higher rank affine special subgroup W_T of a finite rank Coxeter system (W, S) is contained in a nonhyperoctahedral highest virtually Abelian subgroup of W.*

Proof. Let $T_0 := T$. If $T_0^\perp$ is nonspherical, choose a minimal nonspherical subset $T_1 \subset T_0^\perp$. Continue inductively, choosing a minimal nonspherical subset T_{j+1} of $\bigcap_{i=0}^j T_i^\perp$ until the intersection is spherical. This process stops in finitely many steps since S is finite and the nested intersections are strictly decreasing in cardinality. Let $U := T_0 \sqcup \cdots \sqcup T_m$, so that the W_{T_i} are pairwise commuting and $U^\perp$ is spherical.

For each i, if W_{T_i} is irreducible affine, set $B_i := W_{T_i}$. Otherwise, minimality requires that W_{T_i} has no proper nonspherical parabolic subgroups, so if W_{T_i} is not itself irreducible affine then it has no wide parabolic subgroups, so it is hyperbolic, by Moussong's Theorem. Every nonspherical Coxeter group contains infinite dihedral reflection subgroups. Choose B_i to be such a subgroup of W_{T_i}. Define $A := \prod_{i=0}^m B_i$.

The group A is virtually Abelian: each factor B_i has a finite index translation subgroup Λ_i isomorphic to some $\mathbb{Z}^{n_i}$, where $n_i = 1$ if B_i is infinite dihedral and $n_i > 1$ otherwise. Let $\Lambda := \prod_{i=0}^m \Lambda_i$, so that Λ is a finite index subgroup of A that is isomorphic to $\mathbb{Z}^n$ for $n := \sum_{i=0}^m n_i$.

We claim that A is highest. Let $\Lambda' \leq \Lambda$ be a finite index subgroup, and define $\Lambda_i' := \Lambda' \cap \Lambda_i$. Each Λ_i' has finite index in Λ_i, and $\prod_{i=0}^m \Lambda_i'$ has finite index in Λ'. For each i, $\mathrm{Pc}(\Lambda_i') = W_{T_i}$, since Λ_i' is infinite and W_{T_i} contains no proper nonspherical parabolic subgroups. Thus, $\mathrm{Pc}(\Lambda') = W_U$. By construction W_U is essential, so $Z_W(\Lambda') \leq N_W(W_U) = W_U \times W_U^\perp$.

The projection of $Z_W(\Lambda')$ to each factor W_{T_i} centralizes Λ_i'. If W_{T_i} is affine, the centralizer of Λ_i' in B_i is the full translation subgroup Λ_i. Otherwise, Λ_i' is an infinite cyclic subgroup of the hyperbolic group W_{T_i}, so its centralizer in W_{T_i} is virtually cyclic. Thus, in either case Λ_i' has finite index in $Z_{W_{T_i}}(\Lambda_i')$. Since $W_U^\perp$ is spherical, we conclude that Λ' has finite index in $Z_W(\Lambda')$.

This shows that A is highest, because a finite index subgroup A' of A would give a finite index subgroup $\Lambda' := A' \cap \Lambda$ of Λ. Any free Abelian subgroup of W containing A' centralizes Λ', but we just showed that Λ' has finite index in its centralizer, so the centralizer has the same free Abelian rank as Λ' and Λ.

For each i, set $V_i := \Lambda_i \otimes_{\mathbb{Z}} \mathbb{R}$, and let $F_i \leq \mathrm{GL}(V_i)$ be the point group of B_i. The point representation of A acts on $V := \bigoplus_{i=0}^m V_i$. Its image is $F := \prod_{i=0}^m F_i$, where each F_i acts trivially on V_j for all $j \neq i$. For each i, F_i acts on V_i with no nonzero fixed vectors: for an irreducible affine factor this is its finite reflection representation, and for an infinite dihedral factor it is the sign action on $V_i \cong \mathbb{R}$. Set $K := \prod_{i=1}^m F_i$, with K trivial if $m = 0$. Since K acts trivially on V_0 and has no nonzero fixed vectors in V_i for $i > 0$, we have:

$$V^K = V_0 \oplus \bigoplus_{i=1}^m V_i^{F_i} = V_0 \tag{5}$$

Suppose that A is hyperoctahedral. Choose a basis $e_1, \cdots, e_n$ of V such that F acts by signed permutation matrices on that basis. Consider a K–orbit $\mathcal{O}$ of coordinate lines, and let $E_\mathcal{O}$ be their span in V. Choose e_i such that $\mathbb{R}e_i \in \mathcal{O}$. If $-e_i \in Ke_i$ then K fixes no nonzero vectors in $E_\mathcal{O}$. In this case set $v_\mathcal{O} = 0$. Otherwise, for every $\mathbb{R}e_j \in \mathcal{O}$ exactly one of e_j or $-e_j$ is in Ke_i, and the sum $v_\mathcal{O}$ of the included vectors spans the K–fixed subspace of $E_\mathcal{O}$.

Elements of F_0 act by signed permutation on $\{e_1, \ldots, e_n\}$, so they send coordinate lines to coordinate lines. Since F_0 commutes with K, it maps K–orbits of coordinate lines to K–orbits of coordinate lines. Thus, each $f \in F_0$ sends each $E_\mathcal{O}$ to some $E_{\mathcal{O}'}$, and therefore sends $E_\mathcal{O}^K$ to some $E_{\mathcal{O}'}^K$. In particular, F_0 permutes the subspaces $\mathbb{R}v_\mathcal{O}$, respecting their dimensions. Now, the nonzero $v_\mathcal{O}$ are a basis for the fixed subspace of K in V, but that is exactly V_0, by (5), so F_0 acts by signed permutation on a basis of V_0. This contradicts that $W_T = B_0$ is nonhyperoctahedral. Therefore, A is nonhyperoctahedral. □

Remark 5.5. In the proof of Lemma 5.4, the group Λ is a standard free Abelian subgroup in the sense of Krammer [28]. The group A is constructed in an analogous way, but choosing virtually Abelian factors instead of free Abelian. Azuelos and Hagen made a similar definition of "standard virtually Abelian subgroup" immediately before [4, Corollary 7.11], except that they chose the pieces contained in nonaffine factors to be $\mathbb{Z}$ instead of infinite dihedral. This difference turns out to be critical, even though both choices would yield highest virtually Abelian subgroups in our construction. We need the 1–dimensional B_i to be infinite dihedral so that the corresponding F_i has no nonzero fixed vectors on V_i. Replacing any of these infinite dihedral factors by its translation subgroup makes V^K strictly larger than V_0, and this extra space can change hyperoctahedrality. Consider, for example, W that is the product of the $(3,3,3)$–triangle group and the $(4,4,4)$–triangle group. The latter is hyperbolic with no proper nonspherical parabolic subgroups. Our A is of the form $\widetilde{A}_2 \times \widetilde{A}_1$, which is nonhyperoctahedral, by Example 2.13. Azuelos and Hagen choose A' of the form $\widetilde{A}_2 \times \mathbb{Z}$, which *is* hyperoctahedral, by Example 2.12. Both A and A' are highest virtually Abelian subgroups of W, but of the two, only A witnesses that W is not an HHG.

Proof of Theorem 1.3. The equivalence between the hyperoctahedral parabolic condition and the diagrammatic condition is established by Theorem 2.11.

By Theorem 1.2, if the Coxeter system contains no nonhyperoctahedral irreducible higher rank affine parabolic subgroups then W is an HHG. Every HHG admits a coarsely equivariant coarse median: the standard HHS coarse median constructed in [6, Theorem 7.3] is coarsely equivariant by equivariance of the HHS projections and coarse uniqueness of realization.

Suppose W admits a coarsely equivariant coarse median μ. Suppose that (W,S) has a nonhyperoctahedral irreducible higher rank affine parabolic subgroup. Up to conjugation, we may assume it is a special subgroup. By Lemma 5.4, it is contained in a nonhyperoctahedral highest virtually Abelian subgroup A of W. Moreover, in the proof of Lemma 5.4, we constructed such an A with a finite index free Abelian subgroup $\Lambda \leq A$ that also has finite index in $Z_W(\Lambda)$. Fioravanti, Levcovitz, and Sageev [20, Definition 2.13 and Lemma 2.15] show $Z_W(\Lambda)$ is "quasi-submedian", in the sense that $\mu(Z_W(\Lambda), Z_W(\Lambda), Z_W(\Lambda))$ is at finite Hausdorff distance from $Z_W(\Lambda)$. Since A and $Z_W(\Lambda)$ contain Λ with finite index, they are at finite Hausdorff distance from each other in W. Thus A is also quasi-submedian. Define $\mu' \colon A^3 \to A$ by first restricting μ to A^3 and then choosing a closest point in A. Since A is quasi-submedian, $\mu|_{A^3}$ and μ' are at uniformly bounded distance from one another as maps $A^3 \to W$. Since A is a virtually Abelian subgroup of the CAT(0) group W, it is undistorted, so a word metric on A is comparable to the restriction to A of a word metric on W. It follows that μ' is a coarsely equivariant coarse median on A; the metric compatibility conditions of Definition B.1 and Definition B.2 are satisfied by μ with respect to the W–metric, so by bounded distance between μ and μ' they are satisfied by μ' with respect to the W–metric, thus by undistortion they are satisfied by μ' with respect to the A–metric. The existence of a coarsely equivariant coarse median on A contradicts Theorem B.5, since A was constructed to be nonhyperoctahedral. Thus, if W admits a coarsely equivariant coarse median then (W,S) does not contain a nonhyperoctahedral irreducible higher rank affine parabolic subgroup. □

AI Use

The authors conceived the arguments, developed the proofs, and wrote the entirety of this article. Improvements were made after considering comments from Anthropic's Claude Opus 5.5 on earlier drafts.

Appendix A. Promotion of relative HHS structures

In the main text we produce a relative HHG structure on Coxeter groups, observe that the peripheral domains have coordinate spaces that are themselves HHSs, and then need a promotion theorem that combines these structures to make a full HHS/HHG structure. Special cases of such promotion theorems appear in the literature for relatively hyperbolic groups [6, Theorem 9.1] and graph products [8, Theorem 5.1]. Rather than make a Coxeter-specific special case, in this section we distill a reusable, general HHS statement. The argument follows that of Berlyne and Russell [8], who already note that the bulk of their proof only uses general HHS properties, not the specifics of their graph product setting.

Definition A.1 ([1, Definition 3.2])**.** An HHS has *bounded domain dichotomy* if there exists B such that every bounded coordinate space has diameter at most B.

Definition A.2 ([3, Definition 3.3])**.** Let G be a group. A *G–HHS* is an HHS structure on a Cayley graph of G that has bounded domain dichotomy and satisfies all of the requirements of an HHG, Definition 2.19, except that the G action on the domains with bounded coordinate space is not required to be cofinite.

Theorem A.3 (Partial Berlyne-Russell Promotion)**.** *Let $(X, \mathfrak{S})$ be an HHS structure relative to a set of peripheral domains $\mathfrak{P} \subset \mathfrak{S}$. Suppose that $\mathfrak{P} = \mathfrak{P}_0 \sqcup \mathfrak{P}_1$, and for each $P \in \mathfrak{P}_1$ the coordinate space $\mathcal{C}P$ admits an HHS structure $(\mathcal{C}P, \mathfrak{T}_P)$, and that the HHS parameters are uniformly bounded for all $(\mathcal{C}P, \mathfrak{T}_P)$. Then X admits an HHS structure relative to $\mathfrak{P}_0$.*

Suppose, moreover, that $(G, \mathfrak{S})$ is an HHG structure relative to $\mathfrak{P}$, that the decomposition $\mathfrak{P} = \mathfrak{P}_0 \sqcup \mathfrak{P}_1$ is G–invariant, and that the HHS structures on the peripheral coordinate spaces of $\mathfrak{P}_1$ satisfy the following G–compatibility conditions, where for all $g \in G$ and $P \in \mathfrak{P}_1$ the map $g_P \colon \mathcal{C}P \to \mathcal{C}g.P$ is the isometry from the definition of relative HHG, Definition 2.19.

1. *There is an $\sqsubseteq$ and $\perp$–preserving, cofinite action of G on $\coprod_{P \in \mathfrak{P}_1} \mathfrak{T}_P$ with g sending $\mathfrak{T}_P$ bijectively to $\mathfrak{T}_{g.P}$ by $U \mapsto g.U$.*
2. *For all $g, h \in G$, $P \in \mathfrak{P}_1$, $U, V \in \mathfrak{T}_P$, and $x \in \mathcal{C}P$:*
   - *There is an isometry $g_U \colon \mathcal{C}U \to \mathcal{C}g.U$.*
   - *$(gh)_U = g_{h.U} \circ h_U$*
   - *$\delta_{g.U}(g_U(\pi_U(x)), \pi_{g.U}(g_P(x)))$ is uniformly bounded.*
   - *$(U \sqsubsetneq V) \vee (U \pitchfork V) \implies \delta_{g.V}(g_V(\rho_V^U), \rho_{g.V}^{g.U})$ is uniformly bounded.*

Then the promoted structure making G an HHS relative to $\mathfrak{P}_0$ satisfies all of the requirements of being an HHG relative to $\mathfrak{P}_0$ except possibly that the G–action on domains with bounded coordinate spaces may not be cofinite. In particular, if $\mathfrak{P}_0 = \emptyset$ then the promoted relative HHS structure on G is a G–HHS.

Proof. Define $\mathfrak{S}^+ := (\mathfrak{S} \smallsetminus \mathfrak{P}_1) \cup \coprod_{P \in \mathfrak{P}_1} \mathfrak{T}_P$. The new nesting relation is unchanged within $(\mathfrak{S} \smallsetminus \mathfrak{P}_1)$ and within each $\mathfrak{T}_P$. When $Q \in (\mathfrak{S} \smallsetminus \mathfrak{P}_1)$ and $R \in \mathfrak{T}_P$, define $R \sqsubseteq Q$ if and only if $P \sqsubseteq Q$. Similarly, the new orthogonality relation is unchanged within $(\mathfrak{S} \smallsetminus \mathfrak{P}_1)$ and within each $\mathfrak{T}_P$. If $Q \in (\mathfrak{S} \smallsetminus \mathfrak{P}_1)$ and $R \in \mathfrak{T}_P$, define $Q \perp R$ if and only if $Q \perp P$. If $R_0 \in \mathfrak{T}_{P_0}$ and $R_1 \in \mathfrak{T}_{P_1}$ with $P_0 \neq P_1$, then define $R_0 \perp R_1$ if and only if $P_0 \perp P_1$. Note that domains of $\mathfrak{P}_0$ remain nesting minimal in $\mathfrak{S}^+$. Berlyne and Russell then define projections and relative projections and check, [8, Claims 5.2–5.4], that all of the relative HHS axioms are satisfied except possibly for the Containers Axiom. They call $\mathfrak{S}^+$ a "proto-hierarchy structure". They also say, [8, Lemma 5.5], citing [5, Lemma 1.5], that from the uniform HHS constants one can derive a bound on the maximal cardinality of a set of pairwise orthogonal domains in $\mathfrak{S}^+$. With this condition, their [1, Theorem A.1] adds "dummy" container domains whose coordinate spaces are single points to make a larger structure $\mathfrak{S}^{++}$. By their construction, nesting minimal domains of $\mathfrak{S}^+$ remain nesting minimal in $\mathfrak{S}^{++}$, so the domains of $\mathfrak{P}_0$ are nesting minimal in $\mathfrak{S}^{++}$. Thus, $\mathfrak{S}^{++}$ is an HHS structure relative to $\mathfrak{P}_0$. Moreover, $\mathfrak{S}^{++}$ keeps the original nesting and orthogonality relations between domains from $\mathfrak{S}^+$.

The additional G–compatibility conditions are to guarantee that the coarse equivariance conditions of Definition 2.19, which are satisfied by $\mathfrak{S}$, by hypothesis, also are satisfied by $\mathfrak{S}^{++}$. [8, Claim 5.6] shows that the compatibility conditions imply that $\mathfrak{S}^+$ satisfies the definition of relative HHG, except for the Containers Axiom. [8, Theorem 5.7] says that after the addition of the dummy containers, $\mathfrak{S}^{++}$ is an HHG structure on G, relative to $\mathfrak{P}_0$. However, as noted in [3, Remark 3.4], that theorem has a gap: it does not establish cofiniteness of the group action on the dummy container domains. The definition of "G–HHS" circumvents this problem by not requiring cofiniteness on those domains. Bounded domain dichotomy is clear because the G–action on $\mathfrak{S}^+$ is cofinite and the domains of $\mathfrak{S}^{++} \smallsetminus \mathfrak{S}^+$ have single point coordinate spaces. □

Corollary A.4. *The structure $\mathfrak{S}^{++}$ produced by the second branch of Theorem A.3 by adding dummy container domains to the proto-hierarchy structure $\mathfrak{S}^+$ is a relative HHG structure if and only if G acts cofinitely on the set of pairs $(R, \mathfrak{O})$, where $R \in \mathfrak{S}^+$ and $\mathfrak{O}$ is a nonempty, nonmaximal collection of pairwise orthogonal domains in $\mathfrak{S}^+_R \smallsetminus \{R\}$.*

Proof. Pairs of the form $(R, \mathfrak{O})$ index the dummy container domains added to $\mathfrak{S}^+$ in [1, Theorem A.1]. □

The next result gives a sufficient condition that lets us generate G–compatible HHS structures on the peripheral coordinate spaces. Given a G–invariant decomposition $\mathfrak{P} = \mathfrak{P}_0 \sqcup \mathfrak{P}_1$, the idea is to pick a representative P of each G–orbit in $\mathfrak{P}_1$, take an HHS structure $\mathfrak{T}_P$ on $\mathcal{C}P$, and use the G–action to transport that HHS structure to the coordinate spaces of the other peripherals in the G–orbit of P. The desired

G–compatibility will follow, by construction, but for this G–transport to be well-defined we need conditions that say that $\mathrm{Stab}_G(P)$ respects the $\mathfrak{T}_P$ structure.

Proposition A.5. *Suppose $(G,\mathfrak{S})$ is a relative HHG structure with peripherals $\mathfrak{P}$, and suppose there is a G–invariant decomposition $\mathfrak{P}=\mathfrak{P}_0\sqcup\mathfrak{P}_1$. Let $\mathfrak{Q}\subset\mathfrak{P}_1$ be a set of representatives for the G–orbits in $\mathfrak{P}_1$. Let $(\mathcal{C}P,\mathfrak{T}_P)$ be HHS structures for each $P\in\mathfrak{Q}$. Suppose, for all $P\in\mathfrak{Q}$, $g,h\in\mathrm{Stab}_G(P)$, and $U,V\in\mathfrak{T}_P$:*

1. *There is a cofinite action of $\mathrm{Stab}_G(P)$ on $\mathfrak{T}_P$ by bijections $\mathfrak{T}_P\to\mathfrak{T}_P:U\mapsto g.U$ that preserve the $\sqsubseteq$ and $\perp$ relations.*
2. *There are isometries $g_U\colon \mathcal{C}U\to\mathcal{C}g.U$.*
3. *$(gh)_U=g_{h.U}\circ h_U$*
4. *$g_U\circ\pi_U\asymp\pi_{g.U}\circ g_P$*
5. *$(U\sqsubsetneq V)\vee(U\pitchfork V)\implies g_V(\rho^U_V)\asymp\rho^{g.U}_{g.V}$*

Suppose, moreover, that the coarse equivalences of the last two items are uniform. Then HHS structures for the coordinate spaces of all $P\in\mathfrak{P}_1$ can be defined by transporting the chosen structures from $\mathfrak{Q}$ by the G–action, and this collection of HHS structures on the peripheral coordinate spaces satisfies the hypotheses of Theorem A.3, so $\mathfrak{S}^{++}$ satisfies all of the requirements of being an HHG relative to $\mathfrak{P}_0$, except possibly for cofiniteness of the G–action on domains with bounded coordinate spaces. In particular, if $\mathfrak{P}_0=\emptyset$ then $\mathfrak{S}^{++}$ is a G–HHS.

Proof. Suppose $P\in\mathfrak{Q}$. Define the transported HHS structures on the G–orbit of P by:

$$G\times_{\mathrm{Stab}_G(P)}\mathfrak{T}_P:=(G\times\mathfrak{T}_P)/(g,U)\sim(gh,h^{-1}.U)\text{ for }h\in\mathrm{Stab}_G(P)$$

This collection contains the original $\mathfrak{T}_P$ via the identification $U=[1,U]$. The G–action on this collection of domains is $a.[g,U]=[ag,U]$. Define the set $\mathfrak{T}_{g.P}$ of domains of $\mathcal{C}(g.P)$ to be $\{[g,U]\mid U\in\mathfrak{T}_P\}$. This only depends on $g.P$, since if $g.P=k.P$ then $h^{-1}:=k^{-1}g\in\mathrm{Stab}_G(P)$, so:

$$\mathfrak{T}_{k.P}=\mathfrak{T}_{(gh).P}=\{[gh,U]\mid U\in\mathfrak{T}_P\}=\{[g,U']\mid U'\in h.(\mathfrak{T}_P)\}=\{[g,U']\mid U'\in\mathfrak{T}_P\}=\mathfrak{T}_{g.P}$$

For $[g,U],[g,V]\in\mathfrak{T}_{g.P}$ define the nesting and orthogonality relations to be the same as the corresponding relations for $U,V\in\mathfrak{T}_P$. These relations are well-defined and independent of choice of representatives because the action of $\mathrm{Stab}_G(P)$ on $\mathfrak{T}_P$ preserves nesting and orthogonality. Thus, Condition (1) of Theorem A.3 is satisfied. Cofiniteness is also built in, since $\mathfrak{Q}$ is finite and the action of $\mathrm{Stab}_G(P)$ on $\mathfrak{T}_P$ was assumed to be cofinite for each $P\in\mathfrak{Q}$.

For $[g,U]\in\mathfrak{T}_{g.P}$, pick a preferred representative $(g,U)\in[g,U]$, with the convention that for a domain of the form $[1,U]$ we choose $(1,U)$ to be the preferred representative. For a domain of the form $[1,U]$ define its coordinate space to be $\mathcal{C}[1,U]:=\mathcal{C}U$. For $h\in\mathrm{Stab}_G(P)$ define $h_{[1,U]}:=h_U$. For a domain $[g,U]$ with preferred representative (g,U) with $g\neq1$, which implies $g\notin\mathrm{Stab}_G(P)$, define $\mathcal{C}[g,U]$ to be a copy of $\mathcal{C}[1,U]$ with the identification $g_{[1,U]}\colon\mathcal{C}[1,U]\to\mathcal{C}[g,U]$.

For any $k\in G$ and domain $[g,U]$ with preferred representative (g,U), we have $k.[g,U]=[kg,U]$, which has some preferred representative $(kgh,h^{-1}.U)$, for a unique $h\in\mathrm{Stab}_G(P)$. Define the isometry $k_{[g,U]}$ as a composition of the isometries of the previous paragraph (in particular, if $g=1$ this definition agrees with that of the previous paragraph):

$$k_{[g,U]}\colon\mathcal{C}[g,U]\to\mathcal{C}(k.[g,U])=\mathcal{C}[kgh,h^{-1}.U]:=(kgh)_{[1,h^{-1}.U]}\circ(h^{-1})_{[1,U]}\circ(g_{[1,U]})^{-1}$$

This definition satisfies the first two bullets of Condition (2) of Theorem A.3.

Define projection maps by:

$$\pi_{[g,U]}\colon\mathcal{C}(g.P)\to\mathcal{C}[g,U]:x\mapsto\bigcup_{h\in\mathrm{Stab}_G(P)}(gh)_{[1,h^{-1}.U]}\circ\pi_{h^{-1}.U}\circ((gh)_P)^{-1}(x)$$

To see that this projection is coarsely well-defined, we show, using the coarse equivalence of hypothesis (4), that the uniformly bounded sets of the union are coarsely equivalent, so their union is uniformly bounded:

$$\begin{aligned}(gh)_{[1,h^{-1}.U]}\circ\pi_{h^{-1}.U}\circ((gh)_P)^{-1}&=g_{[1,U]}\circ h_{[1,h^{-1}.U]}\circ\pi_{h^{-1}.U}\circ(h_P)^{-1}\circ(g_P)^{-1}\\&=g_{[1,U]}\circ h_{h^{-1}.U}\circ\pi_{h^{-1}.U}\circ(h_P)^{-1}\circ(g_P)^{-1}\\&\asymp g_{[1,U]}\circ\pi_U\circ h_P\circ(h_P)^{-1}\circ(g_P)^{-1}\\&=g_{[1,U]}\circ\pi_U\circ(g_P)^{-1}\end{aligned}$$

We further claim these definitions imply the third bullet of Theorem A.3(2). Specifically, for $[g,U] \in \mathfrak{T}_{g.P}$, $x \in \mathcal{C}(g.P)$, and $k \in G$, we claim:

$$\delta_{k.[g,U]}(k_{[g,U]}(\pi_{[g,U]}(x)), \pi_{k.[g,U]}(k_{g.P}(x))) = 0$$

This is because the two sets intersect, as follows. If $(g,U) \in [g,U]$ and $(kgh_0, h_0^{-1}.U) \in [kgh_0, h_0^{-1}.U] = k.[g,U]$, where $h_0 \in \mathrm{Stab}_G(P)$, are the preferred representatives, we observe from the definitions of the isometries:

$$(gh_0)_{[1,h_0^{-1}.U]} = g_{[1,U]} \circ ((h_0^{-1})_{[1,U]})^{-1}$$

Using this, we produce a nontrivial intersection by specializing each union to the h_0 term:

$$\begin{aligned}
k_{[g,U]}(\pi_{[g,U]}(x)) &= (kgh_0)_{[1,h_0^{-1}.U]} \circ (h_0^{-1})_{[1,U]} \circ (g_{[1,U]})^{-1} \left(\bigcup_{h \in \mathrm{Stab}_G(P)} (gh)_{[1,h^{-1}.U]} \circ \pi_{h^{-1}.U} \circ ((gh)_P)^{-1}(x) \right) \\
&\supset (kgh_0)_{[1,h_0^{-1}.U]} \circ (h_0^{-1})_{[1,U]} \circ (g_{[1,U]})^{-1} \circ (gh_0)_{[1,h_0^{-1}.U]} \circ \pi_{h_0^{-1}.U} \circ ((gh_0)_P)^{-1}(x) \\
&= (kgh_0)_{[1,h_0^{-1}.U]} \circ \pi_{h_0^{-1}.U} \circ ((gh_0)_P)^{-1}(x) \\
&= (kgh_0)_{[1,h_0^{-1}.U]} \circ \pi_{h_0^{-1}.U} \circ ((kgh_0)_P)^{-1} \circ k_{g.P}(x) \\
&\subset \bigcup_{h \in \mathrm{Stab}_G(P)} (kgh)_{[1,h^{-1}.U]} \circ \pi_{h^{-1}.U} \circ ((kgh)_P)^{-1}(k_{g.P}(x)) \\
&= \pi_{k.[g,U]}(k_{g.P}(x))
\end{aligned}$$

Similarly, if $U, V \in \mathfrak{T}_P$ with $U \sqsubsetneq V$ or $U \pitchfork V$, define the relative projection by:

$$\rho^{[g,U]}_{[g,V]} := \bigcup_{h \in \mathrm{Stab}_G(P)} (gh)_{[1,h^{-1}.V]}(\rho^{h^{-1}.U}_{h^{-1}.V})$$

Again, this is a union of uniformly bounded diameter, uniformly coarsely equivalent sets, so it has uniformly bounded diameter: for $h \in \mathrm{Stab}_G(P)$, the pairs $(gh, h^{-1}.U)$ and $(gh, h^{-1}.V)$ are different representatives of $[g,U]$ and $[g,V]$, and, using hypothesis (5):

$$\begin{aligned}
(gh)_{[1,h^{-1}.V]}(\rho^{h^{-1}.U}_{h^{-1}.V}) &= g_{[1,V]} \circ h_{[1,h^{-1}.V]}(\rho^{h^{-1}.U}_{h^{-1}.V}) \\
&= g_{[1,V]} \circ h_{h^{-1}.V}(\rho^{h^{-1}.U}_{h^{-1}.V}) \\
&\asymp g_{[1,V]}(\rho^U_V)
\end{aligned}$$

Finally, the fourth bullet of Theorem A.3(2) follows, because again we claim that for any $P \in \mathfrak{Q}$, $g, k \in G$ and $[g,U], [g,V] \in \mathfrak{T}_{g.P}$ with $[g,U] \sqsubsetneq [g,V]$ or $[g,U] \pitchfork [g,V]$ we have:

$$\delta_{k.[g,V]}(k_{[g,V]}(\rho^{[g,U]}_{[g,V]}), \rho^{k.[g,U]}_{k.[g,V]}) = 0$$

Again, this is because the two sets intersect. Let (g,V) and $(kgh_0, h_0^{-1}.V)$ be the preferred representatives of $[g,V]$ and $k.[g,V] = [kgh_0, h_0^{-1}.V]$. Specialize the unions defining $\rho^{k.[g,U]}_{k.[g,V]} = \rho^{[kg,U]}_{[kg,V]}$ and $\rho^{[g,U]}_{[g,V]}$ to their h_0 terms. Then the identity $(gh_0)_{[1,h_0^{-1}.V]} = g_{[1,V]} \circ ((h_0^{-1})_{[1,V]})^{-1}$ and the definition of $k_{[g,V]}$, as in the previous calculation, show that both $\rho^{k.[g,U]}_{k.[g,V]}$ and $k_{[g,V]}(\rho^{[g,U]}_{[g,V]})$ contain $(kgh_0)_{[1,h_0^{-1}.V]}(\rho^{h_0^{-1}.U}_{h_0^{-1}.V})$. □

Appendix B. Coarsely equivariant coarse medians on virtually Abelian groups

Recall that the proof of Theorem 1.3 uses Lemma 5.4 to escalate a nonhyperoctahedral irreducible higher rank affine parabolic subgroup to a nonhyperoctahedral highest virtually Abelian subgroup. In this section we show that for finitely generated virtually Abelian groups hyperoctahedrality is equivalent to the existence of a coarsely equivariant coarse median.

Definition B.1. A *C–coarse median space* (X, d, μ) is a metric space (X, d) equipped with a ternary operation $\mu\colon X^3 \to X$ such that, writing $x \sim_C y$ if $d(x,y) \le C$, the following hold for all $v, w, x, y, z \in X$:

(C1) $\mu(x,x,y) = x$
(C2) $\mu(x,y,z) = \mu(y,x,z) = \mu(x,z,y)$
(C3) $d(\mu(x,y,z), \mu(x,y,w)) \le Cd(z,w) + C$

(C4) $\mu(\mu(x,w,y),w,z) \sim_C \mu(x,w,\mu(y,w,z))$
(C5) $\mu(v,w,\mu(x,y,z)) \sim_C \mu(\mu(v,w,x),\mu(v,w,y),z)$

It is a *coarse median space* if it is a C–coarse median space for some C.

This is the normalized formulation of Niblo, Wright, and Zhang [34, Theorem 3.1], which is equivalent to Bowditch's formulation [11] up to a uniformly bounded change of the operation. Such a change allows (C1) and (C2) to hold exactly [34, Remark 2.10]. By [34, Proposition 3.11], given (C1), (C2), and (C3), each of (C4) and (C5) implies the other, after possibly increasing C, so it suffices to take either one in the definition.

A coarse median can be pulled back by a quasiisometry, with a uniformly bounded adjustment to enforce (C1) and (C2) if necessary. Thus, the existence of a coarse median is a quasiisometry invariant. In particular, it makes sense to say that a finitely generated group is a *coarse median group* if one, equivalently every, Cayley graph with respect to a finite generating set is a coarse median space.

Definition B.2. Let G be a finitely generated group equipped with a word metric d and a C–coarse median μ. We call μ *coarsely equivariant* if for all $g,x,y,z \in G$:

(C6) $g\mu(x,y,z) \sim_C \mu(gx,gy,gz)$

Definition B.3. A *cubulation* of a group G is a proper action $G \curvearrowright Y$ by cubical automorphisms on a CAT(0) cube complex.

Definition B.4 ([20, Section 2.4])**.** A coarse median μ on a finitely generated group G is *cubical* if there exist a cocompact cubulation $G \curvearrowright Y$, a vertex $y \in Y^{(0)}$, and a constant $D \geq 0$ such that, for all $f,g,h \in G$, if m_Y is the median on $Y^{(0)}$:

$$d_Y\big(\mu(f,g,h)y, m_Y(fy,gy,hy)\big) \leq D$$

Equivalently, μ is at uniformly bounded distance from the pullback of m_Y by the orbit quasiisometry $g \mapsto gy$.

Theorem B.5. *Let G be virtually $\mathbb{Z}^n$, for some $n \geq 1$. The following are equivalent:*

(1) G is hyperoctahedral.
(2) G is cocompactly cubulable.
(3) G admits a cubical coarse median.
(4) G admits a coarsely equivariant coarse median.

The main work for us is the implication (4) $\implies$ (1). The idea is to use two limiting arguments: first an argument of Bowditch turns a coarsely equivariant coarse median on G into an equivariant topological median on an asymptotic cone of G, which is homeomorphic to $\mathbb{R}^n$ (see [19, Chapter 10] for background on asymptotic cones). Then we use arguments modeled on those of Haettel, [21, Propositions G and 4.1] to say that there is a biLipschitz median embedding of a standard n–cube into the asymptotic cone. A differentiation and rescaling argument produces a median on the cone that is F–equivariant and linearly conjugate to the standard n–cube median. The second part is broken off separately as an auxiliary lemma:

Lemma B.6. *For $n \geq 1$, let $F \leq \mathrm{GL}_n(\mathbb{R})$ be finite. Suppose μ is a Lipschitz median on $\mathbb{R}^n$ that is equivariant with respect to the actions of F and the translations of $\mathbb{R}^n$. Then F is conjugate into $\mathrm{O}_n(\mathbb{Z})$.*

Proof. Since μ is Lipschitz it is, in particular, continuous, so [9, Corollary 3.10] implies that it is locally convex with rank at most n. By [21, Corollary 3.7], there does not exist a Lipschitz, locally convex median of rank strictly less than n on $\mathbb{R}^n$, so the rank of μ is exactly n. Thus, [21, Proposition G] says there exists a biLipschitz median embedding $\phi\colon ([0,1]^n,\nu) \to (\mathbb{R}^n,\mu)$, where ν is the standard coordinate median on $\mathbb{R}^n$. The goal now is to transfer F–equivariance from μ to ν.

By Rademacher's theorem, ϕ is differentiable almost everywhere. Choose an interior point p of the n–cube at which ϕ is differentiable, and let $D_p\phi$ be the derivative of ϕ at p, which is invertible, since ϕ is biLipschitz. For all $u \in \mathbb{R}^n$ and all sufficiently small $\epsilon > 0$ the point $p + u\epsilon$ is in the cube, so we can define $\phi_\epsilon(u) := (\phi(p+\epsilon u) - \phi(p))/\epsilon$, and we have $\phi_\epsilon(u) \xrightarrow{\epsilon\to 0} (D_p\phi)(u)$. For $\epsilon > 0$, define $\mu_\epsilon(x,y,z) := \mu(\epsilon x, \epsilon y, \epsilon z)/\epsilon$.

Translation equivariance of μ and median preservation of ϕ give, for all $u,v,w \in \mathbb{R}^n$ and small enough ϵ:

$$\begin{aligned}
\epsilon\mu_\epsilon(\phi_\epsilon(u),\phi_\epsilon(v),\phi_\epsilon(w)) &= \mu\left(\phi(p+\epsilon u)-\phi(p), \phi(p+\epsilon v)-\phi(p), \phi(p+\epsilon w)-\phi(p)\right) \\
&= \mu(\phi(p+\epsilon u),\phi(p+\epsilon v),\phi(p+\epsilon w)) - \phi(p) \\
&= \phi(\nu(p+\epsilon u, p+\epsilon v, p+\epsilon w)) - \phi(p)
\end{aligned}$$

$$\begin{aligned}&= \phi(p + \epsilon\nu(u,v,w)) - \phi(p)\\ &= \epsilon\phi_\epsilon(\nu(u,v,w))\end{aligned}$$

Since the μ_ϵ share the same Lipschitz constant as μ, the family is equicontinuous, so:

$$\begin{aligned}(D_p\phi)(\nu(u,v,w)) &= \lim_{\epsilon\to 0} \phi_\epsilon(\nu(u,v,w))\\ &= \lim_{\epsilon\to 0} \mu_\epsilon(\phi_\epsilon(u), \phi_\epsilon(v), \phi_\epsilon(w))\\ &= \lim_{\epsilon\to 0} \mu_\epsilon\left((D_p\phi)(u), (D_p\phi)(v), (D_p\phi)(w)\right)\end{aligned}$$

Thus, we can define a limit median:

$$\mu_0(a,b,c) := \lim_{\epsilon\to 0} \mu_\epsilon(a,b,c) = (D_p\phi)\left(\nu\left((D_p\phi)^{-1}(a), (D_p\phi)^{-1}(b), (D_p\phi)^{-1}(c)\right)\right)$$

Since μ is equivariant with respect to each linear $f \in F$, so is each μ_ϵ, and hence so is μ_0. Thus, ν is $(D_p\phi)^{-1}F(D_p\phi)$–equivariant.

Since each $f' \in (D_p\phi)^{-1}F(D_p\phi)$ is linear it fixes 0. By equivariance, it takes the median interval $[0, e_i]_\nu$ to the median interval $[0, f'e_i]_\nu$. Since f' is a linear automorphism and $[0, e_i]_\nu$ is a nondegenerate line segment, so is $[0, f'e_i]_\nu$, which implies $f'e_i$ has exactly 1 nonzero coordinate. Since f' is invertible, this implies that f' permutes the coordinate lines. By averaging an inner product on $\mathbb{R}^n$ over the action of the finite group $(D_p\phi)^{-1}F(D_p\phi)$ and normalizing the coordinate vectors to have unit length with respect to this new inner product, conclude that $(D_p\phi)^{-1}F(D_p\phi)$ permutes these coordinate vectors up to sign. □

Proof of Theorem B.5. Hagen [23, Theorem B] proved (1) $\iff$ (2) when G is crystallographic. We argue that the virtually Abelian case of (1) $\implies$ (2) reduces to the crystallographic case. As above, let A be a finite index normal $\mathbb{Z}^n$ subgroup of G, let $V := A \otimes_{\mathbb{Z}} \mathbb{R} \cong \mathbb{R}^n$, and let $\rho\colon G \to \mathrm{GL}(V)$ be the representation induced by conjugation on A, with $F := \rho(G)$. Let $Q := G/A$, which is a finite group.

A classical construction of Zassenhaus relates virtually $\mathbb{Z}^n$ and crystallographic groups. This is well-known, compare [25, Section 3.2] or [4, Section 2.13], but we will re-use the following notation in a later part of the proof. The construction in the proof of [40, Satz 1] gives a map $\tau\colon G \to V$ and a homomorphism $\alpha\colon G \to V \rtimes F$ such that:

$$\alpha : g \mapsto (\tau(g), \rho(g))$$

Furthermore, the restriction of α to A is $\alpha|_A\colon a \mapsto (a, \mathrm{I}_n)$. Since α is a homomorphism:

$$\tau(gh) = \tau(g) + \rho(g)(\tau(h)) \tag{6}$$

Together with the A–normalization, this gives, for $a \in A$ and $h \in G$:

$$\tau(ah) = a + \tau(h) \tag{7}$$

Define an affine action of G on V by:

$$g.x := \tau(g) + \rho(g)(x)$$

By averaging over the finite F–action, we can construct an F–invariant inner product on V that makes this action isometric. Observe that A is a uniform lattice of translations. Since A has finite index in G, the G–action is geometric. The kernel N of the G–action is finite, since it intersects each of the finitely many cosets of A in at most one element. Zassenhaus assumes that the conjugation action of Q on A is faithful and concludes that the affine action is faithful, so G is crystallographic, but the same argument applies to say that G/N is crystallographic. Moreover, the point group of G/N is again F, acting on V by the original representation ρ. Hence, if G is hyperoctahedral then so is G/N. In that case, [23, Theorem B] establishes that G/N acts geometrically by cubical automorphisms on a CAT(0) cube complex. Since N is finite, there is a geometric, cubical G–action on the same cube complex, by factoring through G/N. This proves (1) $\implies$ (2).

Pulling back the cube complex median gives a cubical coarse median on G. This proves (2) $\implies$ (3).

An orbit map is equivariant, and cubical automorphisms preserve the standard cubical median. Pulling that median back by a quasiinverse to the orbit map is therefore coarsely equivariant, proving (3) $\implies$ (4).

Now suppose μ is a coarsely equivariant coarse median on G. Fix a nonprincipal ultrafilter ω and scaling sequence $r_i \to \infty$. Take the constant sequence at the identity element 1_G of G to be the observation sequence. There is an extended pseudometric on sequences (g_i) and (h_i) in G defined by $\lim_\omega d_G(g_i, h_i)/r_i$. The asymptotic cone of G with respect to $(\omega, (r_i), (1_G))$ is the metric quotient of the sequences at finite distance

from (1_G). Since G is virtually $\mathbb{Z}^n$, this cone is biLipschitz equivalent to $\mathbb{R}^n$. Moreover, we can make an explicit biLipschitz equivalence to V using the maps α, τ, and ρ from the first part of the proof. First observe that $g.0 = \tau(g)$, so $\tau\colon G \to V$ is the orbit map based at $0 \in V$. Since the action is geometric, τ is a quasiisometry. Define a map from the asymptotic cone to V by:

$$[(g_i)] \mapsto \lim_\omega \frac{\tau(g_i)}{r_i}$$

The scaling kills off the additive quasiisometry constant of τ, so this defines a biLipschitz map from the cone to its image. It is surjective because for every $v \in V$ and r_i there is an $a_i \in A$ at uniformly bounded distance from $r_i v$, so $a_i/r_i \to v$, which implies that the point $[(a_i)]$ of the cone maps to v. By [11, Proposition 9.1], μ induces a Lipschitz median μ^ω on the cone, and thus on V.

To apply Lemma B.6, we must verify that μ^ω is equivariant with respect to translations and F. Given $v \in V$, choose $a_i \in A$ with $a_i/r_i \to v$. Since $\tau(a_i g_i) = a_i + \tau(g_i)$, by (7), the sequence of left translations by a_i induces translation by v on the cone. The median equivariance error for a_i is the constant C of (C6), so the median equivariance error for v is $\lim_\omega C/r_i = 0$. Thus, μ^ω is translation equivariant. For $g \in G$, by (6):

$$\frac{\tau(gg_i)}{r_i} = \frac{\tau(g)}{r_i} + \rho(g)\left(\frac{\tau(g_i)}{r_i}\right)$$

Thus, left multiplication by g induces the linear map $\rho(g)$ on the cone. The same error estimate as in the translation case implies that μ^ω is $\rho(g)$–equivariant. Since $F = \rho(G)$, conclude that μ^ω is F–equivariant. Now Lemma B.6 says that F is conjugate into $\mathrm{O}_n(\mathbb{Z})$, so G is hyperoctahedral. This proves (4) $\Longrightarrow$ (1). □

Corollary B.7. *The irreducible affine Coxeter groups of types $\widetilde{A}_2$, $\widetilde{A}_n$ for $n \geq 4$, $\widetilde{E}_6$, $\widetilde{E}_7$, $\widetilde{E}_8$, $\widetilde{F}_4$, and $\widetilde{G}_2$ admit no coarsely equivariant coarse medians.*

Proof. Apply Theorem B.5 and Theorem 2.11. □

There are other results in the literature building off of Hagen's equivalence between hyperoctahedrality and cocompact cubulation that give additional equivalent conditions, either for crystallographic groups or more generally for virtually Abelian groups. These include a result of Petyt and Spriano [38, Theorem 4.4], who add "being an HHG" for crystallographic groups, and Hoda [25, Theorem 7.1], who adds "being Helly" and "being coarsely injective" for virtually Abelian groups. Theorem B.5 adds to these results the condition of "admitting a coarsely equivariant coarse median". Hoda's proof also runs through the asymptotic cone, so in that sense our arguments are similar.

We note an imprecision in [38, Theorem 4.4]: their goal is to show that for crystallographic groups being an HHG, being cocompactly cubulated, and being hyperoctahedral are equivalent, but in place of hyperoctahedrality they use that "the point group embeds into the hyperoctahedral group $\mathrm{O}_n(\mathbb{Z})$". This is strictly weaker than hyperoctahedrality, recall Remark 2.14, so their invocation of [23, Theorem B] is not fully justified. [4, Corollary 7.9] closes this gap. Alternatively, one can complete their proof by observing that an HHG admits a coarsely equivariant coarse median and applying Theorem B.5. Establishing the required linear conjugacy, rather than just an abstract embedding of the point group, is precisely the purpose of the differentiation and rescaling argument in the proof of Lemma B.6.

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